



Microsoft New Future of Work Report 2025

A summary of recent research from Microsoft and around the world that can help us create a new and better future of work with AI.

Welcome to the 2025 Microsoft New Future of Work Report!

As you sit down to read the 2025 New Future of Work report, it's worth pausing to consider the thread that ties the past five years of reports together. The inaugural New Future of Work report, published in 2021, focused on new ways people could work without relying on colocation as a key productivity tool. The second, in 2022, centered on the reintroduction of physical offices and the emergence of hybrid work. In 2023, we explored how large language models could reshape everyday work, and, in 2024, how those advances moved from promise to real-world impact.

Each year, as I've written this introduction, I've found myself saying that the previous year marked a once-in-a-lifetime generational shift. But after five years, it's clear that the reports aren't capturing a series of separate revolutions. Rather, they are chapters in a single story of the digital evolution of collaboration, each representing a phase that builds on, and is enabled by, what came before.

Last year's report highlighted research showing that AI delivers substantial gains in individual productivity. The next frontier, covered in this year's report, is collective productivity: how teams, organizations, and communities can get better together. AI can bridge gaps of time, distance, and scale, but only if built correctly. We must design AI to support shared goals, group context, and the norms of collaboration, and this requires not just new tools but new ways of working.

Microsoft's mission to empower every person and every organization on the planet to achieve more remains a stable north star as the terrain shifts. If the past five years taught us anything, it's that the future of work is not something that happens to us, it's something we create together, as a research community, as an industry, and as a public. As always, we invite you to join that effort, approaching it with curiosity, intentionality, and guided by evidence, so the next chapter of work is better for everyone.

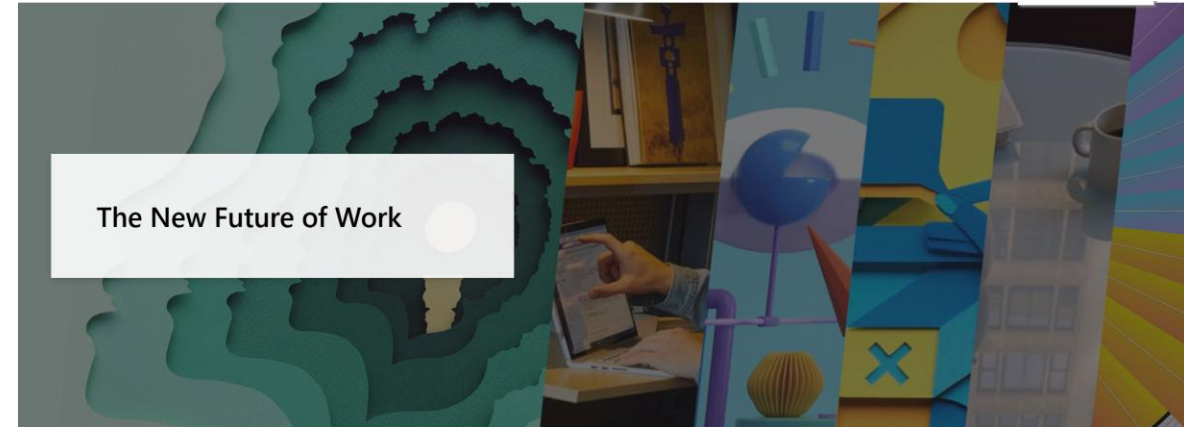
– Jaime Teevan, Chief Scientist and Technical Fellow

This report is a product of Microsoft's New Future of Work Initiative

Microsoft has shaped information work for over 50 years and the New Future of Work (NFW) initiative has helped it navigate the past five years of profound transformation. While the initiative was born out of the COVID-19 pandemic and subsequent shifts to remote and hybrid work, since 2023 it has focused on the incorporation and growing role of tools powered by large AI models.

Throughout this transformative period, the NFW initiative has brought together researchers from across Microsoft to conduct primary research and synthesize existing findings from the literature. The goal is not merely to predict what changes are coming, but to actively create a new future of work that is equitable, inclusive, meaningful, and productive.

The fifth edition of the annual NFW report represents another year of research and analysis, adding to the growing body of knowledge on AI and work. The evidence and insights presented here are not monolithic and represent contributing researchers' perspectives rather than Microsoft's corporate views. Additional research papers, practical guides, and white papers are available at: aka.ms/nfw.



[Overview](#) [Workstreams](#) [Publications](#) [Videos](#) [News & features](#)

The New Future of Work is a cross-company research initiative dedicated to creating solutions for a future of work that is meaningful, productive, and equitable. It began during the pandemic in response to an urgent need [to understand remote work practices](#). When many people returned to the office, the focus shifted to [supporting the hybrid work transition](#). Then in 2023, another generational shift in work occurred when language [models made the leap](#) from the lab into the real world, a shift that could make the changes to remote and hybrid work look small by comparison.

The future of work with AI is not a forgone conclusion, and this initiative exists to not just study work with AI, but to help Microsoft build a new future of work with AI that empowers every person on the planet.

This site features research from the initiative that has been published in peer-reviewed scientific venues, as well as resources to help you navigate a rapidly changing work environment and thrive in the age of AI. We **recently published our 2024 Report** that summarizes some of the exciting work in this space.

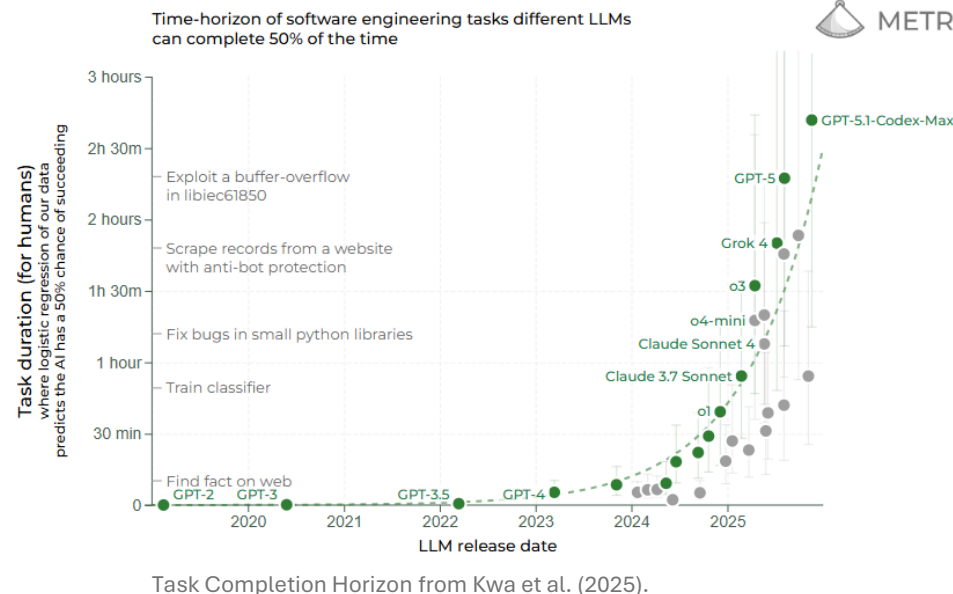


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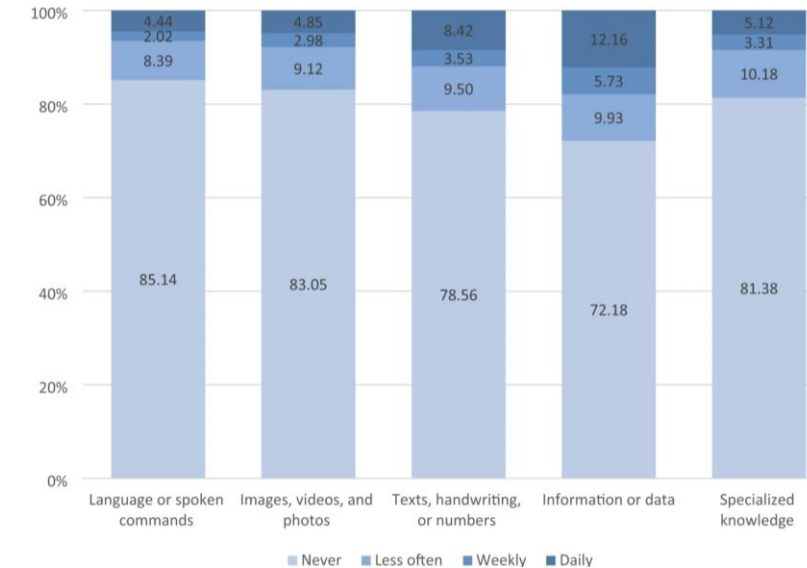
Key context: AI capabilities continue to advance, especially due to reinforcement learning

- Long-horizon task completion capability is measurably accelerating. METR's 50% task-completion time horizon shows frontier agents' reliable task length has been rising exponentially with an ~ 7 -month doubling time, turning "agent progress" into a concrete capability trend (Kwa et al., 2025).
- Verifiable reinforcement learning (RL) post-training (rewarding correct, checkable outcomes) enabled strong gains on hard math/coding style tasks even when starting from a base model with no labeled reasoning traces. (DeepSeek-AI et al., 2025).
- Scalable test-time compute frameworks gained traction, with open-weight models achieving IOI 2025 gold-level performance, showing repeatable "more compute $\rightarrow$ higher score" curves in competitive programming. (Samadi et al., 2025).
- Multi-turn RL for tool-using/search agents is now beating prompt-only baselines by learning from experience over longer action horizons; on a legal document search benchmark, an RL-trained 14B agent reports 85% vs 78% against frontier-class models, with additional gains when allowed more turns. (Kalyan & Andrews, 2025).
- Multi-model routing + aggregation is shifting from one-shot "pick a model" to an RL-trained sequential policy that can think, call multiple models, and integrate responses while explicitly optimizing performance–cost tradeoffs—improving results across multi-hop QA style evaluations. (Zhang et al., 2025).



Investment in and adoption of generative AI continues to grow

- Effects of and gains from generative AI will be mediated by adoption (Jaffe et al., 2024).
- In 2024, Generative AI had \$33.9 billion in global private investment—an 18.7% increase from 2023. Public investment has also grown (Maslej et al., 2025).
- AI use at work has been increasing, but heterogeneous:
 - Enterprise ChatGPT messages increased 8x in the past year (Chatterji et al., 2025).
 - A German survey found 38% of employed respondents used AI for work (Giering & Kirchner, 2025).
 - A survey of enterprise leaders found that usage and confidence were highest in IT & Procurement and lowest in Marketing/Sales and Operations; the Tech/Telecom, Professional Services, and Finance industries are leading (Korst et al., 2025).
 - A large 2024 survey of Americans found men were more likely than women (29.1% vs 23.5%) to use generative AI for work (Bick et al., 2024).
- On the consumer side, in June 2025, ChatGPT had over 700 million weekly active users globally (Chatterji et al., 2025).
 - The gender gap (based on first names) of consumer ChatGPT users is gone, a dramatic change since early 2023 when >80% were male (Chatterji et al., 2025).



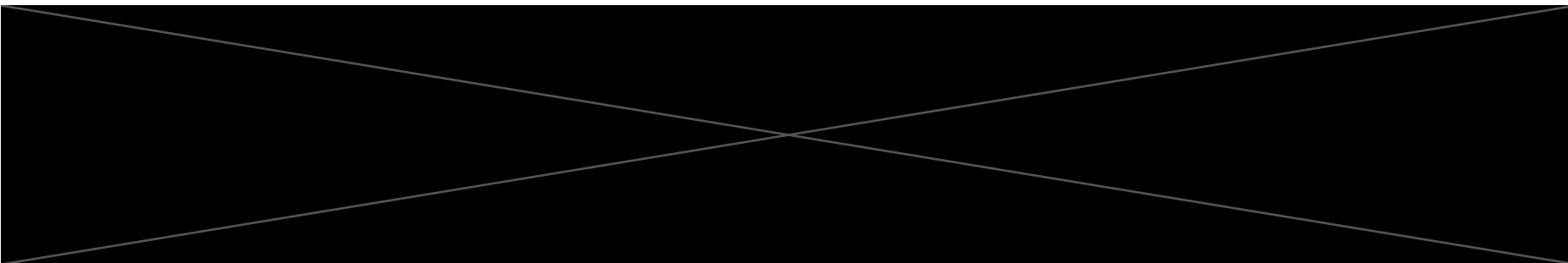
Use of AI-based systems at work to recognize and process (weighted percentages) from Giering & Kirchner (2025).

Organizational AI adoption depends on employees as much as leaders

- Across industries, the intention to use AI is influenced by social norms learned from leaders and peers (Kelly et al., 2023).
- Workers can be reluctant to adopt top-down mandated AI products that prioritize efficiency above quality and creativity, undermining the traditional view of humans as the core value driver of businesses. This reluctance limits the success of AI pilot programs (Young et al., 2025; Sharma, 2025; Murire, 2024).
- Leaders can facilitate AI adoption through clear communication supporting AI use, demonstrating their own learning, and setting realistic expectations about what AI can accomplish (Carter et al., 2024; Tursunbayeva & Chalutz-Ben Gal, 2024).
- AI products that integrate human thinking, creativity, and expertise while amplifying their value can promote adoption without raising concerns about replacement (Ali et al., 2025; Young et al., 2025; Sharma, 2025). For example, an AI assistant can act as a thought partner, helping users explore ideas and connect concepts across their knowledge base.
- Some of the best ways an organization might find to use AI “come from the edge, not the center” (Winsor, 2024). Organizations can facilitate AI adoption by creating systems and incentives for employees to share how they use AI with one another (Tursunbayeva & Chalutz-Ben Gal, 2024; Winsor, 2024).
- Employees are more likely to experiment with using AI and to share those insights with others when they feel safe and trust their organizations (Tursunbayeva & Chalutz-Ben Gal, 2024; Bankins et al., 2021).
- Many employees, particularly Gen X, will not adopt tools that make them conform to a way of working - they want products that are flexible enough to fit personal ways of working (Rozsa et al., 2023; Doblinger, 2023).

Centering worker voice in AI design boosts productivity, satisfaction, and skill growth—driving both business success and worker flourishing

- Worker involvement in technology design promotes sustainable productivity and job satisfaction. Historical and contemporary research consistently shows that when workers' expertise and perspectives inform the design and deployment of workplace technologies, organizations achieve more sustainable improvements in productivity and well-being (Trist & Bamforth, 1951; Roethlisberger & Dickson, 1939; Hackman & Oldham, 1976).
- Ethnographic and HCI research demonstrates that workers adapt technology in creative ways, and that participatory design—where workers are co-designers—results in tools that better fit real workflows and foster higher adoption (Suchman, 1987; Orr, 1996; Awumey et al., 2024; Ehn, 1993; Doellgast et al., 2025).
- Combining technical and social science research methods can create AI systems that improve worker skills and satisfaction—not just accuracy—by embedding human-centric metrics, workers' values, and skill-building into their design (Bucinka, 2025).
- Data-driven workplace monitoring (telemetry) has mixed effects and should be governed with worker input for best outcomes. While monitoring and algorithmic management can boost short-term output, they often increase stress and erode trust unless workers help define what is measured and how data is used (Pentland, 2012; Wells et al., 2007; Ajunwa, 2023).

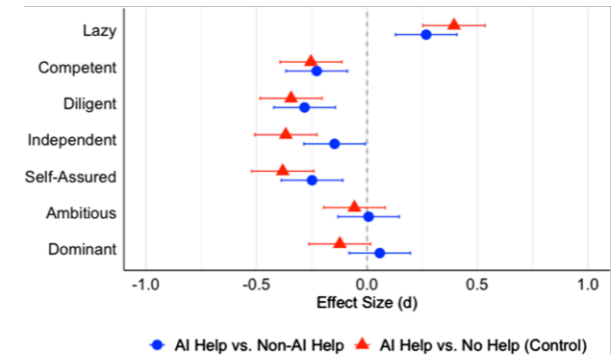


CEOs expect AI to transform their businesses, but leading organizational AI adoption can be challenging

- A 2025 IBM survey of 2000 CEOs in 33 countries and 24 industries found that most CEOs expect AI to transform their businesses. Other industry research shows leaders believe having the most advanced generative AI is crucial to remaining competitive (de Bellefonds et al., 2024; IBM Institute for Business Value, 2025).
- However, organizational leaders have difficulties developing top-down AI strategies for many reasons, including the rapid diffusion of AI technologies, the speed with which they change, the need to communicate and reach alignment about AI, the need to prioritize AI against other concerns, and the challenge of reimagining workflows and processes (de Bellefonds et al., 2024; Leonardi, 2023).
- A comparative case study of AI adoption in the Dutch public sector found adoption barriers included organizational inflexibility and risk intolerance, as well as structural separation between exploration and exploitation. For example, a data science team might be put in charge of exploring AI without operational alignment or frontline and leadership support to exploit and scale their ideas (Selten & Klievink, 2024).
- Chen & Tajdini (2024) surveyed managers involved in Marketing and AI at US-based firms involved in high-tech consumer goods, industrial equipment, and financial services. They found that organizational intensity of AI adoption was driven by top management support, customer orientation, and emerging industry social norms toward adopting AI (Chen & Tajdini, 2024).
- Additionally, organizations are best able to adopt AI when they are innovative, experimental and learning-oriented, supportive, and collaborative (de Bellefonds et al., 2024; Tursunbayeva & Chalutz-Ben Gal, 2024, Sternfels & Atsmon, 2025).

Social norms shape how people interpret others' (possible) AI use, often with negative consequences

- Research has shown that suspected use of AI can result in moral evaluations. People who use AI assistance expect to be, and often are, evaluated as “lazier”, less competent, less diligent, less trustworthy, and less moral (Reif et al., 2025; Schilke & Reimann, 2025; Zhou et al., 2025). These effects are lessened when evaluators, including managers, have experience with and positive attitudes towards AI (Reif et al., 2025; Schilke & Reimann, 2025).
- Being seen as using AI can harm relationships with colleagues, perhaps because AI-using employees may be seen as “slacking” rather than putting in personal effort (Zhou et al., 2025). Paradoxically, the act of disclosing the use of AI can erode trust (Schilke & Reimann, 2025).
- The extent to which these “perceptual harms” (Kadoma et al., 2025) vary across social identities is unclear:
 - Reif et al. (2025) found no gender differences in experiments across occupational categories.
 - Kadoma et al. (2025) found that male (and East Asian) freelancers were more likely to be suspected of using AI and hence evaluated negatively.
 - Software engineers who used AI received lower competency ratings on identical work, and this effect was doubled for females, with women receiving a competency score 13% less for identical code (vs 6% for men) (Gai et al., 2025).



Differences in evaluations for AI help vs. Non-AI help and AI help vs. Control. Positive d values indicate higher values in the AI Help condition while negative d values indicate lower values in the AI Help condition. Error bars represent 95% CI. From Reif et al. (2025)

The Responsible AI Organizational Maturity Model provides a map for organizational change

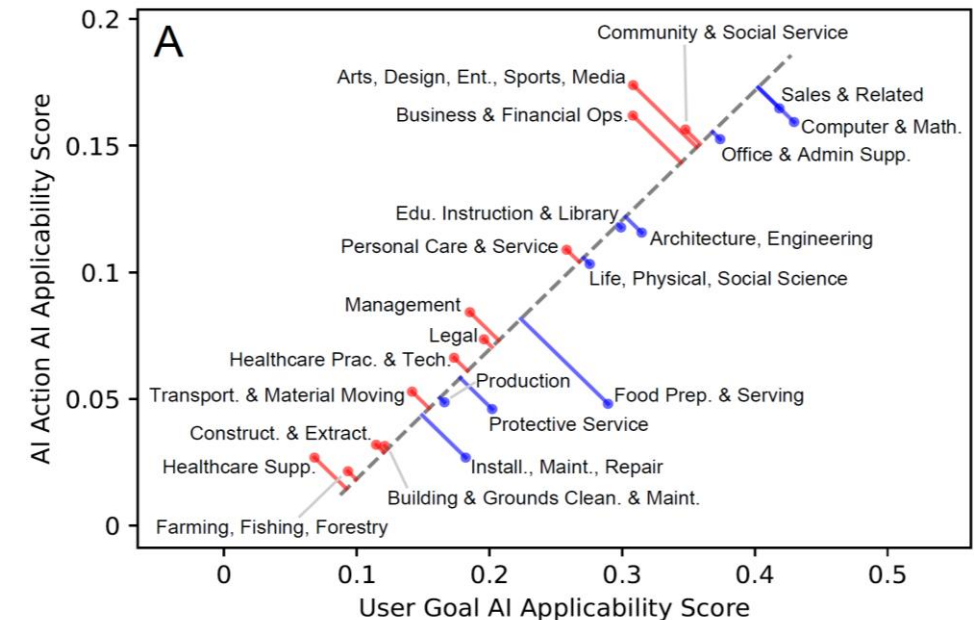
- AI adoption and capability building require organizational transformation (Kemp, 2024). Maturity models are useful tools for guiding organizational transformation. Existing maturity models focus on building AI capability (Alsheiabni et al., 2019; Vaish et al., 2021) and AI adoption (Hansen et al., 2024).
- The Responsible AI Organizational Maturity Model (RAI-OMM) provides a map for organizations to advance their responsible AI strategy and practice (Heger et al., 2025). The RAI-OMM is forward-looking and best used for planning, not evaluation.
- Based on interviews and co-design sessions with 90 RAI experts and practitioners, the RAI-OMM identifies 24 aspects (dimensions) that need to be considered for mature RAI practice and describes, for each dimension, five distinct maturity levels.
- RAI maturity requires leadership investment, aligned organizational practices, and holistic change management strategies that address both technological and human dimensions (Duran, 2025; Shekshnia, 2025; Wang et al., 2025; Herrmann & Pfeiffer, 2023; Yunusa, 2025).
- RAI maturity dimensions are interdependent and fall into three categories: Organizational Foundations require leadership commitment and investment in organization-wide infrastructure; Team Approach dimensions highlight the necessity of cross-discipline collaboration; finally, these make possible mature RAI Practice, which is characterized by deep integration into AI development & deployment processes.



The 24 dimensions in the RAI-OMM fall into three main categories (Heger et al., 2025).

Analyses of LLM logs are showing for which activities the tools are used, and which occupations do those activities

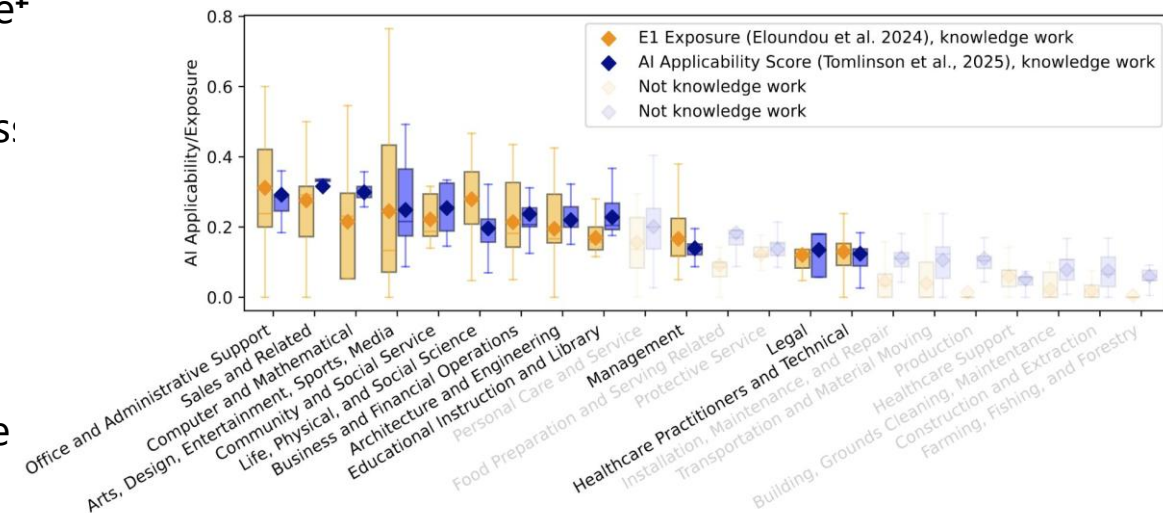
- Analysis of ChatGPT found growth in work-related messages but even faster growth in non-work-related messages.
 - “Practical Guidance,” “Seeking Information,” and “Writing” are the three most common topics, accounting for ~80% of all use messages (Chatterji et al., 2025).
- Researchers at Anthropic found that 37% of Claude usage was for tasks associated with Computer and Mathematical occupations (Handa et al., 2025).
- An analysis of Microsoft Bing Copilot logs looked separately, at the work activities associated with users' goals and the AI actions (Tomlinson et al., 2025).
 - The most common user goals involved learning and communicating; AI actions were largely communicating and explaining; both sides had lots of writing activities.
 - Aggregating activities to occupations, most occupations have some AI-applicable tasks, with information workers including sales, computer occupations, media, and administrative occupations at the top.



Graph shows average AI applicability for the user goal (x-axis) and the AI action (y-axis) (Tomlinson et al., 2025).

AI could broaden access to high-value knowledge work outputs and opportunities, if societal and technical barriers can be overcome

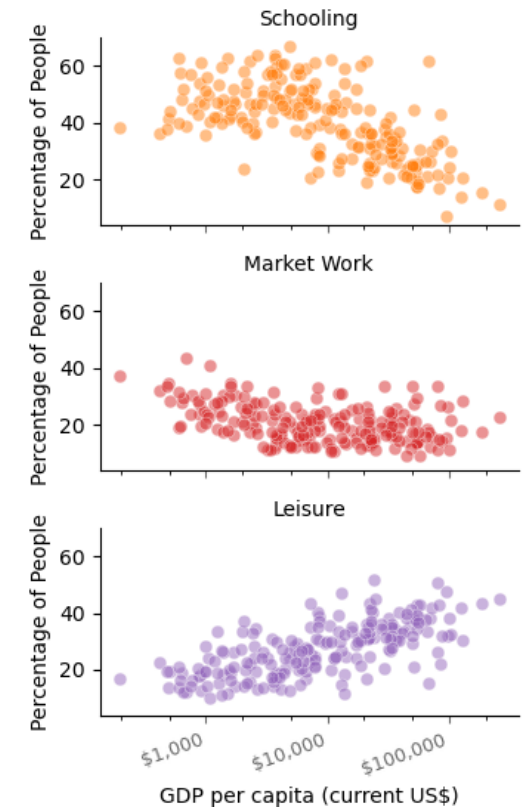
- AI can increase people's efficiency in multiple domains of knowledge work (Dillon et al., 2025; Brynjolfsson et al., 2025; Cui et al., 2024).
- These capabilities could broaden the set of people who have access to knowledge work skills and outputs (Autor, 2024); this is bolstered by evidence that AI can shrink skill gaps in knowledge work (Brynjolfsson et al., 2025; Cui et al., 2024). Autor argues this could allow more people to engage in high-barrier-to-entry work, like medical or legal decision-making.
- For AI to be democratizing in this way, technical improvements are needed to address the "rich-get-richer" effect in open-ended, creative tasks and policy innovation will be needed to mitigate emerging divides in AI investment, access, and benefits (Daepf et al., 2025; Microsoft AI Economy Institute, 2025).



Estimates of how applicable AI is to different occupational categories, from Eloundou et al. (2024) and Tomlinson et al. (2025). Knowledge work occupations are highlighted. Boxplots are employment-weighted and diamonds show means. Both studies show AI to be most useful for knowledge work occupational tasks (Daepf et al., 2025).

AI use in low-income countries is growing, particularly for schooling

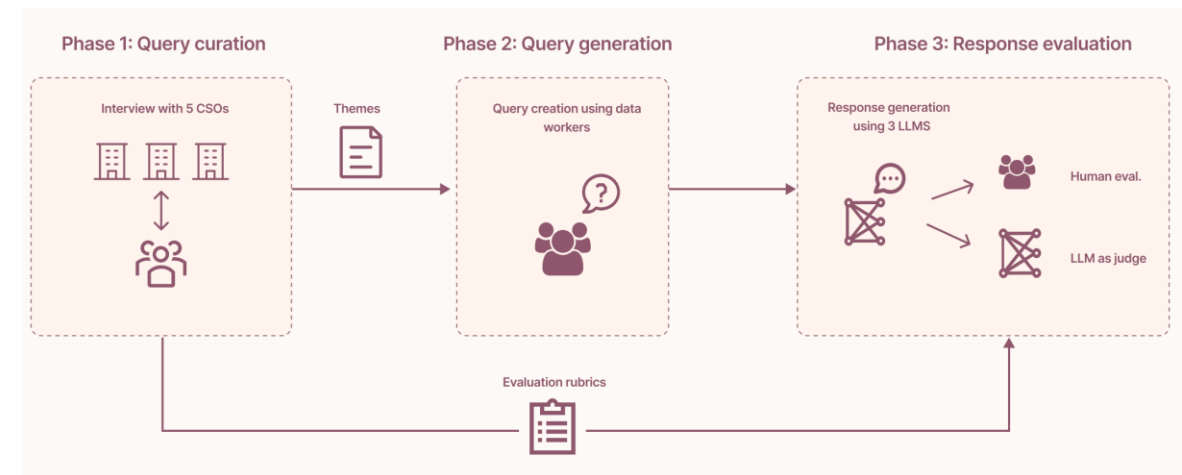
- Though AI usage remains highest in high-income and technologically advanced countries (Microsoft AI Economy Institute, 2025; Appel et al., 2025), the last year has seen dramatic usage growth in low- to middle-income countries, narrowing the gap (Chatterji et al., 2025).
- In one survey, people in Asia and Latin America were more likely to agree that “products and services using AI have more benefits than drawbacks” (e.g., China: 83%, Mexico: 70%) while agreement is lower in Europe and the Anglosphere (e.g., US: 39%, Netherlands: 36%) (Maslej et al., 2025).
- Adoption is highest in countries that have invested in digital infrastructure and education, and where the predominant language is one well-served by existing models (Microsoft AI Economy Institute, 2025).
 - When local languages are not well-served by AI, people sometimes use English instead: chats are conducted in English at disproportionate rates relative to the fraction of the population that speaks English in African and Asian countries, but not in Europe or the Americas (Slaughter & Daepp, in prep.).
- How AI is used also differs across place. In a study of early chatbot adopters, LLM usage for school increases with GDP per capita, while use for leisure decreases (see chart; Slaughter & Daepp, in prep.). This may be partially due to differences in the share of school-age population or in the amount of leisure time.



Patterns in Usage Domain among Early Adopters of Bing Copilot, the percentage of users in a country who predominantly had conversations in the specified domain (schooling, market work, or leisure) by GDP per capita (Slaughter & Daepp, in prep.).

Synthetic data and representative evaluations are advancing LLMs for diverse languages and cultures

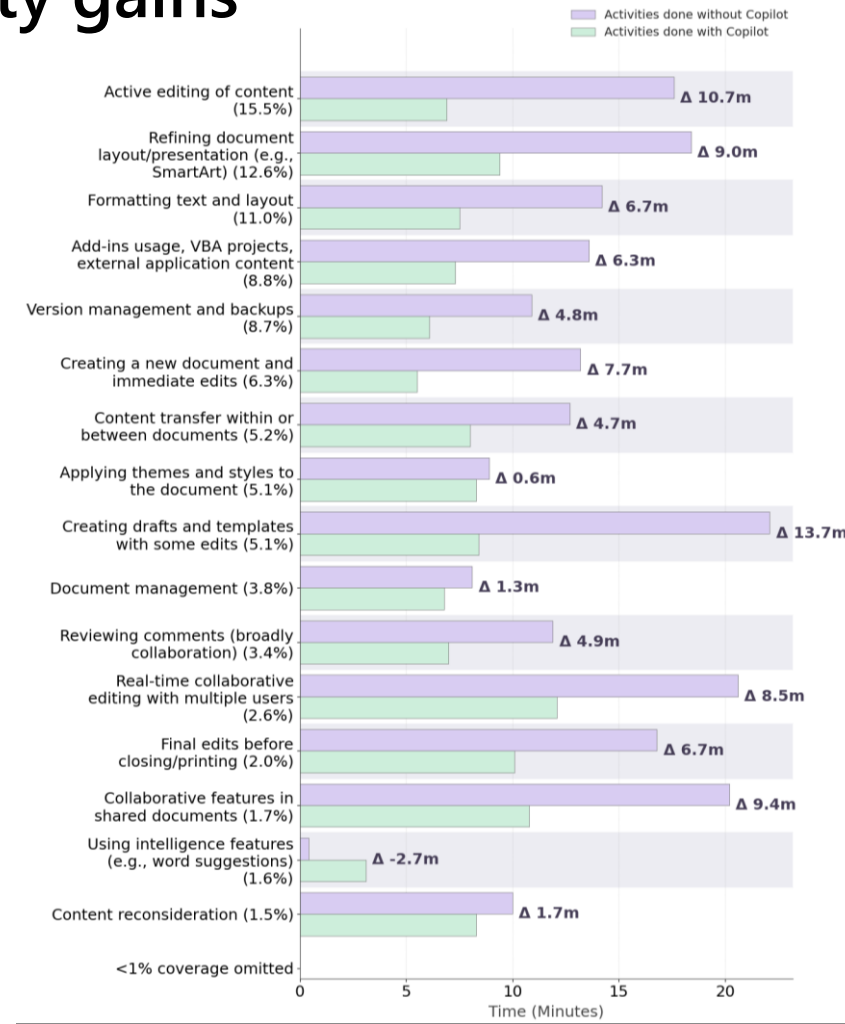
- Thoughtfully constructed, culturally-grounded synthetic data has shown promise in reducing the data gap between high-resource and under-resourced languages (Guduru et al., 2025). The Updesh project (Chitale et al., 2025) provides a framework for culturally-grounded multilingual synthetic data and a case study on Indian languages shows improved performance of models fine-tuned on this data.
- Coverage, representativeness, scalability, and trust continue to pose significant challenges in multilingual and multicultural evaluation. The Samiksha project (Hamna et al., 2025) addresses these issues by developing benchmarks from the ground up by incorporating inputs from a wide range of stakeholders and community members.
 - Samiksha aims to establish a realistic and representative benchmark for Indian languages that reflects key use cases.
 - Insights from such assessments can contribute to building models that perform effectively in various languages and cultural contexts.



The Samiksha benchmark creation and evaluation pipeline. Feedback on what to evaluate is provided by Civil Society Organizations (CSOs), data workers expand themes into data points in their own languages, adding cultural context when appropriate; responses from LLMs are evaluated using human evaluation and LLM-judges (Hamna et al., 2025).

AI use is associated with time savings and productivity gains

- Surveyed ChatGPT Enterprise users attribute 40–60 minutes saved per day from use of AI (Chatterji et al., 2025). The savings are heterogeneous. LLM-based estimates of time savings from Claude usage suggested they varied by occupation and task—for example, 80–85% for legal and management tasks, but only 20% for checking diagnostic images (Tamkin & McCrory, 2025).
- OpenAI designed 1,320 tasks to mimic the work product of predominantly digital occupations in high-value sectors. Frontier LLMs approached quality parity with human experts: the win+tie shares for the top model ranged from 33-56% across industries and tie rates were low (Patwardhan et al., 2025).
- To see how time spent on tasks varied with Copilot usage, researchers developed *WorkflowView*, which uses an LLM to categorize telemetry action sequences into high-level workflow activities (Verma & Counts, 2025).
 - Privacy-preserving analysis of one month of telemetry of 50k Copilot-enabled Word users shows an average difference of 7 minutes per accepted Copilot output.
 - Copilot use is associated with a difference of 10.7 minutes in editing of content and 0.6 minutes in applying themes and styles. These variations can guide more effective integration of AI tools in productivity workflows.



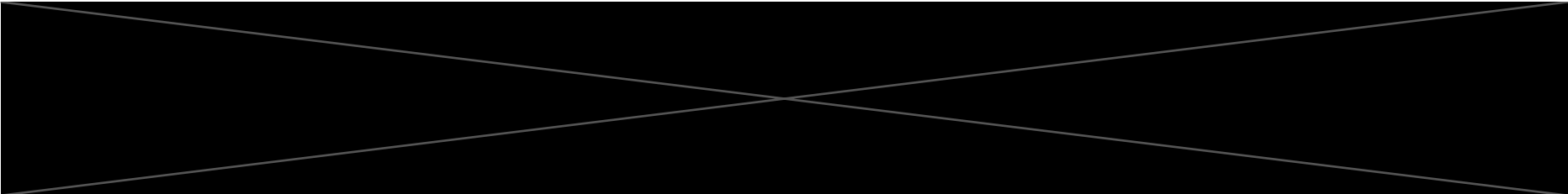
Average time spent by *WorkflowView* Activity and Copilot usage (Verma & Counts, 2025)

The rise of AI “workslop” has productivity risks and market effects

- AI “workslop” refers to AI-generated work content that appears useful but lacks substance, is incomplete, or contains inaccuracies. Such content undermines productivity by forcing recipients to interpret, correct, or redo the work (Niederhoffer et al., 2025; Madsen & Puyt, 2025). Workslop may be a key reason why individual productivity gains are not seen at the group or organizational level.
- In Niederhoffer et al.’s (2025) survey of 1,150 U.S. employees, 40% received workslop in the past month, estimated at 15% of content. Most slop flows between peers (40%), but it also moves upward (18%) and downward (16%) in hierarchies.
- Workslop is part of the broader generative AI “slop” phenomenon, which is reshaping markets by flooding them with low-cost and low-quality content (Miklian & Hoelscher, 2025; Tullis, 2025; Pendergrass et al., 2025).
- Technical solutions are still nascent. One approach to metrics focuses on judging information utility, information quality, and style quality (Shaib et al., 2025), but this would need to be combined with accuracy checks (e.g. MAD-Fact by Ning et al., 2025), ideally with access to internal data or document repositories.
- Employee training about awareness of AI limitations and critical evaluation skills can reduce workslop by helping people identify and correct low-value outputs before they enter workflows (Park et al., 2025; Simkute et al., 2024).

Issue	Definition	Consequence
Volume	Scale and abundance of generative output	Crowding out human creativity and visibility
Velocity	Speed of production and circulation	Outpacing fact-checking and moderation
Variety	Range of forms, genres, and modalities	Expansion into all cultural and knowledge domains
Value	Erosion of cultural and epistemic worth	Devaluation of originality and meaning
Verification	Problems of truth, trust, and reliability	Epistemic pollution and misinformation
Visibility	Algorithmic amplification and ranking	Rewarding slop via platform incentives
Virality	Meme-like diffusion and contagious spread	Rapid uptake and normalization of slop

The 7Vs of AI Slop (Adapted from Madsen & Puyt, 2025).

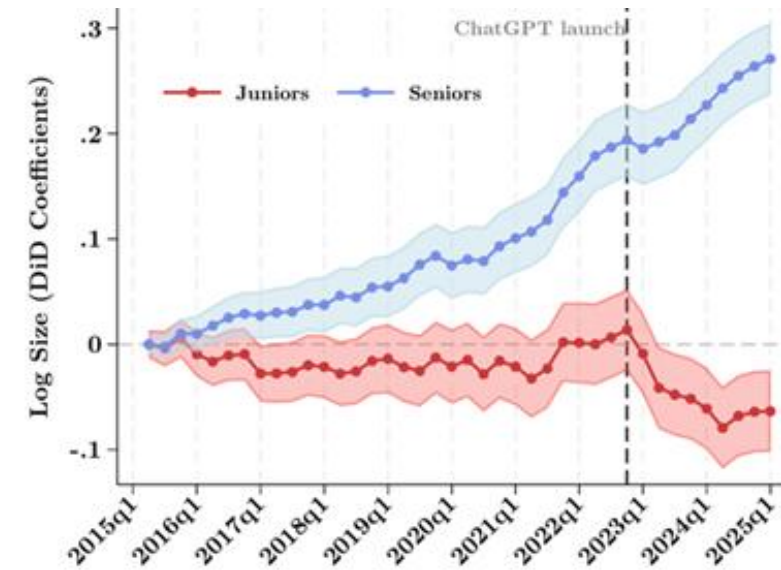


Labor market outcomes cannot be predicted based on existing limited evidence on automation vs augmentation

- Researchers have tried to categorize which tasks are more likely to be augmented or automated by generative AI, either in theory using an LLM classifier (Eloundou et al., 2024) or by analyzing what users are trying to achieve in conversations with LLMs as distinct from what activities the LLM is performing (Tomlinson et al., 2025).
- However, the technical question of what activities can be delegated to the tools versus which activities it can assist with cannot answer the labor market question of which occupations will see increases or decreases in employment or wages.
- Employment and wages depend on how occupations are restructured and whether there is market demand for increased output.
 - When ATMs were invented, tellers' jobs were refactored to focus on other tasks and employment increased.
 - Theory suggests that wages are most likely to increase when the tasks that are automated are the ones requiring less expertise than other activities that are part of that occupation (Autor & Thompson, 2025).
- Researchers have also elicited workers' preferences about which tasks they would prefer to have automated (Shao et al., 2025).
 - Work preferences may predict the tasks for which AI will be adopted. They may also mediate the effect on wages if workers need to be paid more to use AI for some tasks than for others.
 - The top reasons given for wanting automation were freeing up time for high-value work, tasks being repetitive or tedious, and improving the quality of work.
- Evidence on new tasks – that were not being done prior to AI – requires more research (and more time for them to emerge).

Aggregate effects remain small, but early-career employment impacts from AI are emerging

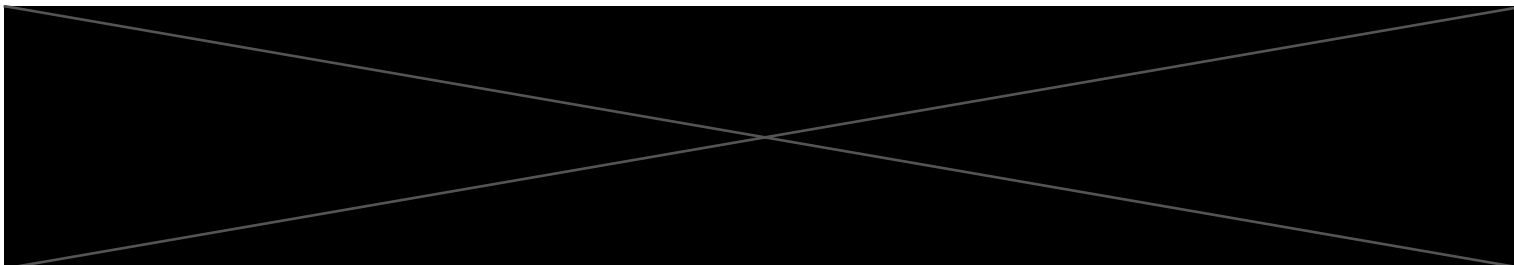
- Large-scale studies in Denmark and the US find no significant effect of AI on unemployment (Chen et al., 2025), working hours (Humlum & Vestergaard, 2025), or job openings (Hartley et al., 2025). However, hiring of AI talent has increased by more than 300% over the past eight years (LinkedIn, 2025).
- For earnings, results vary from slightly increased earnings (Hartley et al., 2025), especially for occupations with high AI exposure (Chen et al., 2025), to no significant effects (e.g., Humlum & Vestergaard, 2025) to reduced salaries in high-wage occupations (Klein Teeselink, 2025).
- There is evidence of negative effects for younger workers, whose roles rely less on tacit experience, making them more vulnerable to automation and less shielded by firm-specific skills. Declines for younger workers might be offset by growth among older workers and less-exposed occupations.
 - Payroll data suggests employment for workers aged 22–25 in highly AI-exposed jobs fell by about 13% compared to less-exposed roles. The authors tested for firm-level shocks, remote work, and sector-specific effects but note they cannot fully rule out other trends influencing exposed occupations (Brynjolfsson et al., 2025).
 - Resume and job posting evidence shows hiring for junior/entry-level roles slows in exposed occupations after firms adopt AI (Hosseini & Lichtinger, 2025; Klein Teeselink, 2025).



Difference between employment at firms adopting and not adopting generative AI, separately for junior and senior employee (Hosseini & Lichtinger, 2025).

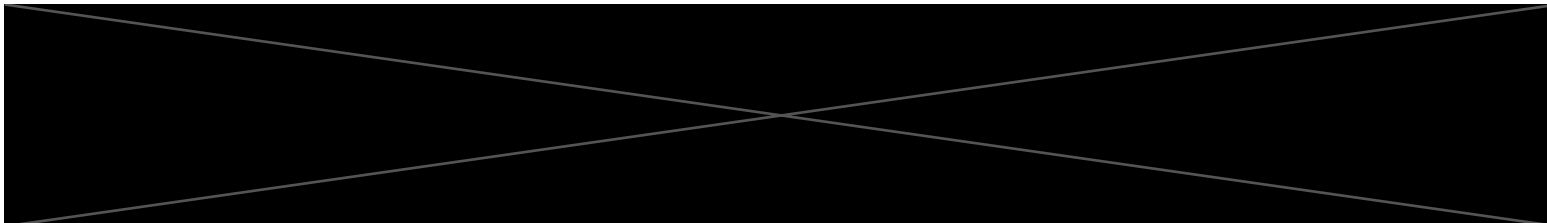
AI adoption is reshaping career paths and skill requirements within occupations

- Recent evidence suggests AI adoption affects career decisions and occupational mobility. Workers using AI chatbots are more likely to switch occupations (Humlum & Vestergaard, 2025), and search intensity for apprenticeships in cognitive and language-intensive fields has declined after chatbot introduction, signaling shifts in career preferences (Goller et al., 2025).
- Worker-level evidence from Germany shows AI exposure changes the mix of activities and required skills inside occupations. Unlike robots, AI reduces non-routine abstract tasks and increases demand for high-level routine tasks like oversight and evaluation (Engberg et al., 2025; Gathmann et al., 2024). AI adoption increases complexity in augmentation-prone roles while reducing skill requirements in automation-prone roles (Chen et al., 2024).
- Roles requiring AI skills are nearly twice as likely to also request analytical thinking, resilience, ethics, or digital literacy. A doubling of AI-specific job postings is associated with roughly 5% higher demand for these complementary skills, while demand for easily substitutable tasks such as basic data skills or translation declines slightly (Mäkelä & Stephany, 2025). Job postings requiring AI skills are growing over 70% year-over-year, extending beyond technical roles (LinkedIn, 2025a; 2025b).
- Workers exposed to AI gain most from retraining focused on broad skills rather than narrow AI-specific roles. Encouragingly, occupations exposed to AI show strong adaptive capacity, suggesting retraining can work if job loss occurs (Hyman et al., 2025; Manning & Aguirre, 2025). Separately, experimental evidence suggests that while generative AI can enable non-technical workers to perform technical tasks, these gains may be temporary and dependent on continued tool use; workers lose the capability to perform those tasks once access ends, indicating no lasting skill development (Wiles et al., 2024).



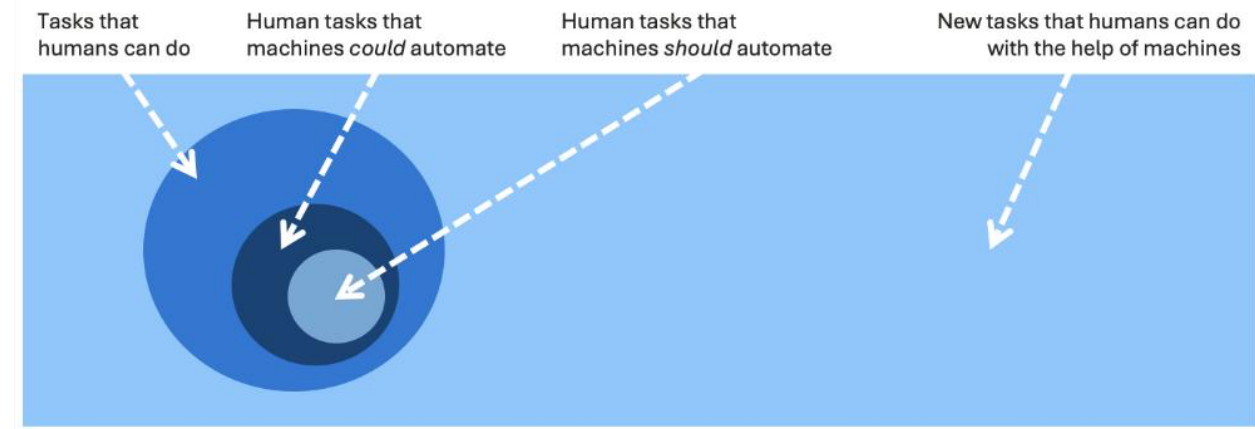
Theory suggests AI shifts the value of work toward human judgment and decision making

- While empirical evidence on AI's long-term impact is still emerging, theoretical work is shaping our understanding of possible future scenarios and the complex ways AI could transform work, expertise and organizations (del Rio-Chanona et al., 2025).
- AI might act as a "bicycle for the mind," boosting output and initially narrowing inequality by automating routine work. As AI advances, human judgment becomes increasingly critical – recognizing improvement opportunities and selecting the right action under ambiguity – areas tied to context, ethics, and creativity where AI still struggles (Agrawal et al., 2025a). Whether this increases inequality between workers depends on how autonomously AI is used and on users' skill levels (Ide & Talamàs, 2025).
- By turning hidden, local knowledge into data that can be shared and analyzed, AI lets companies coordinate on a larger scale (Brynjolfsson & Hitzig, 2025). This may favor bigger organizations and centralized decision-making. However, automating too much – especially entry-level work – risks losing hands-on learning and tacit skills that drive long-term innovation (Ide, 2025).
- As AI gets more powerful, it can automate not only routine but also expert-level tasks, making economic growth and wages increasingly dependent on compute rather than labor. If AI takes over work closer to the existing knowledge frontier and humans retain a comparative advantage mostly on the far frontier, where entirely new knowledge must be created, humans may focus even more on the most creative and novel challenges (Restrepo, 2025; Agrawal et al., 2025b; Celis et al., 2025).
- The increasing power of AI also allows it to automate chains of tasks in a production process. A task where a human has an advantage over AI may nonetheless be automated if it is adjacent to tasks where AI performs well. This can result in non-linear productivity gains that arise when marginal improvements in AI trigger discrete reorganizations of work (Demirer et al., 2025).



Automation tends to cap upside – augmentation and innovation expand opportunity

- A version of this figure has appeared in the last two New Future of Work Reports, but it remains as important (if not more) today.
- Automation-first approaches mainly remove cost from existing tasks, limiting gains to labor arbitrage and replication (Brynjolfsson, 2022; Autor, 2022).
- Strategies focused on augmentation and innovation can create new categories of work and value, driving positive-sum growth. For instance, Amazon's success came from reinventing retail through human-machine complementarity, not automating cashiers.
- Automation also risks reducing the autonomy, recognition, and connection workers receive (making work less “meaningful”), all of which matter for engagement, performance, and wellbeing (Bailey et al., 2019; Allen et al., 1990).
- Automation without augmentation or innovation can also reduce worker bargaining power, thereby concentrating wealth and power and amplifying inequality (Brynjolfsson, 2022).



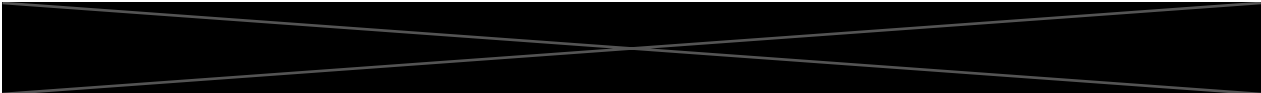
Opportunities for augmenting humans are far greater than opportunities to automate existing tasks (Adapted from Brynjolfsson, 2022).

Establishing common ground is essential for human-AI collaboration (1)

- *Common ground* is the shared knowledge and assumptions that allow people to understand each other and coordinate effectively.
- Building common ground is essential to human communication and is increasingly recognized as important in human-AI interaction.
- Common ground is achieved through a dynamic process called *grounding*: speakers provide ongoing evidence of their understanding or confusion, and work to address gaps in mutual understanding (Clark & Brennan, 1991; Clark, 1996).
- Grounding includes positive evidence of understanding, such as acknowledgments, relevant next conversational turns, and continued attention as well as negative evidence, such as requests for clarification.
- Current LLMs generate language with less conversational grounding than people, instead generating text that suggests that shared understanding has already been reached (Shaikh et al., 2024; see figure for examples).

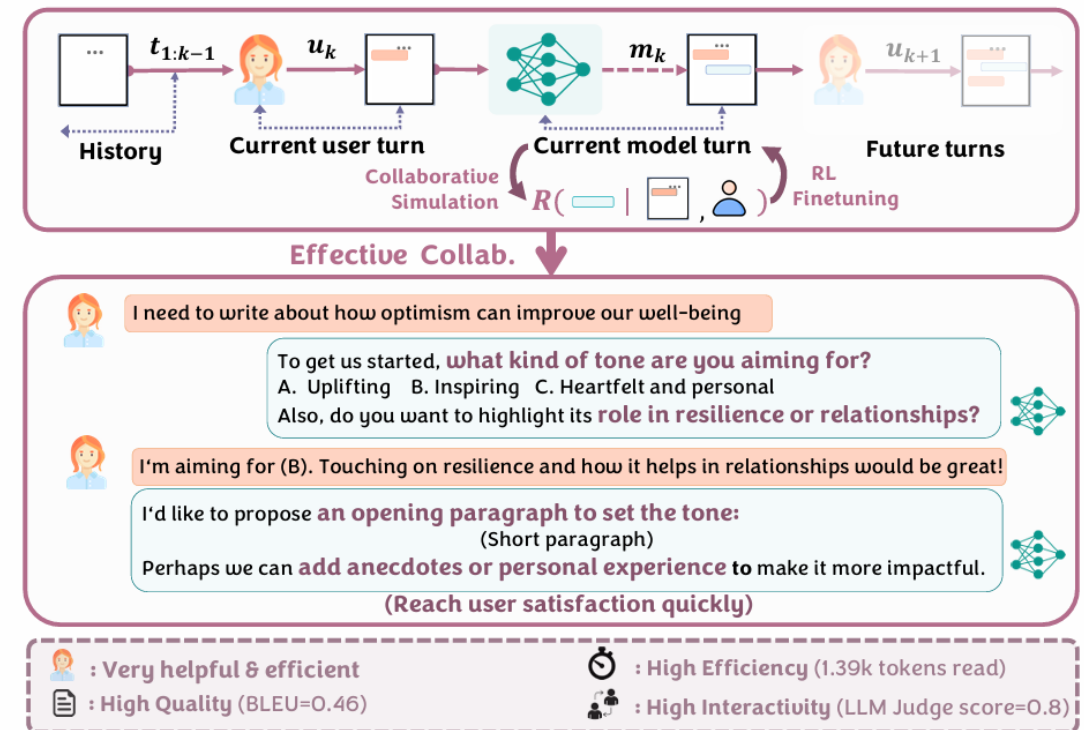
Grounding Act	Example
Clarification <i>TSCC</i>	student: So I am the leader of my class. It is like the president or mayor. teacher: Oh, like a head girl? ↪ GPT: That's great to hear! Being the leader of your class is a big responsibility [...]
Follow-up <i>ESConv</i>	seeker: My friends, who I also work with, are upset with me because I am leaving for a new position. I am sad that they are upset. support: Ah, that can be awkward. Are you happy about the new position? ↪ GPT: I can understand why that would make you feel down... Remember, it's okay to [...].
Acknowledgment <i>Persuasion</i>	persuadee: I [donate] as much as I can, which is admittedly not as much as I would like to. persuader: I know what you mean! Sometimes it is hard to find the extra time or money to help those that need it. ↪ GPT: That's wonderful to hear! [...] Would you be interested in supporting Save the [...]

Examples of grounding acts from Shaikh et al. (2024). GPT-3.5 does not show grounding behaviors in response to the same statements.



Establishing common ground is essential for human-AI collaboration (2)

- Bansal et al. (2024) identify 12 challenges to establishing common ground in human-agent interaction, including general communication challenges (e.g. choosing the appropriate level of detail), challenges with conveying information from a user to an agent (e.g. in relation to goals and preferences), and challenges with conveying information from an agent to a user (e.g. in relation to its capabilities, current and planned actions, and their side effects).
- Tolzin and Janson (2025) identify five mechanisms that could support common ground in human-agent interaction: support for joint action, an underlying knowledge base, mental models, social features, and embodiment.
- Shaikh et al. (2025) study grounding by analyzing LLM interaction logs. They find that early grounding failures predicted later interaction breakdowns. They develop a preliminary intervention aimed at mitigating grounding failures. This prompts the model to ask follow-up questions or request clarifications when predicted to be necessary.
- Other recent advances seek to support the user's overall goal over multiple turns. COLLABLLM (Wu et al., 2025; see figure) results in higher task performance and interactivity, and work by Poelitz and McKenna (2025) improved models' ability to generate clarifications and incorporate user-initiated corrections.



COLLABLLM (Wu et al., 2025) incorporates multi-turn rewards from collaborative simulation, resulting in better task completion and more efficient and interactive conversations.

Human-AI co-creation is shifting from one-shot outputs to context-aware, interaction-driven partnerships

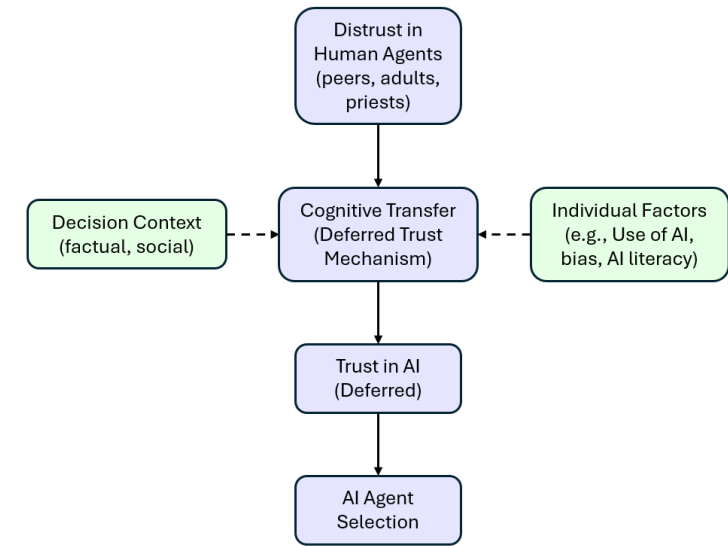
- Modern LLMs have fundamentally shifted content creation workflows from assuming perfect results in a single attempt to engaging in iterative, multi-turn collaborative refinement (Mysore et al., 2025). This transformation reflects both technological advances in LLM capabilities and profound changes in how users perceive and interact with AI (Reza et al., 2025), increasingly viewing it as a creative partner rather than a passive tool (Wan et al., 2024).
- These patterns are emerging consistently across diverse domains—from writing and design (Zhou et al., 2024) to workplace communication (Das et al., in prep.) and software development (Deineha et al., 2025).
- Large-scale empirical analysis of real-world LLM-assisted writing reveals that users rarely accept AI's initial outputs passively. Instead, they engage in complex, multi-turn dialogues characterized by Prototypical Human-AI Collaboration Behaviors (PATHs)—including revising intents, exploring alternatives, asking clarifying questions, and iteratively adjusting style and content (Mysore et al., 2025).
- Recent studies show modern AI enables non-linear collaboration frameworks where humans and AI iteratively diverge (exploring ideas) and converge (refining toward consensus), replacing rigid task delegation. This shift mirrors creative *prototyping*—users experiment, test alternatives, and refine through iterative feedback (Zhou et al., 2024; Yukita et al., 2025).



Users follow up their original requests to collaborate with LLMs in writing sessions. Mysore et al. (2025) identify prototypical human-AI collaboration behaviors (PATHs) and find statistically significant correlations between users' writing intents and PATHs.

Collaborative human-AI decision-making can be made more effective through understanding, trust, and precision

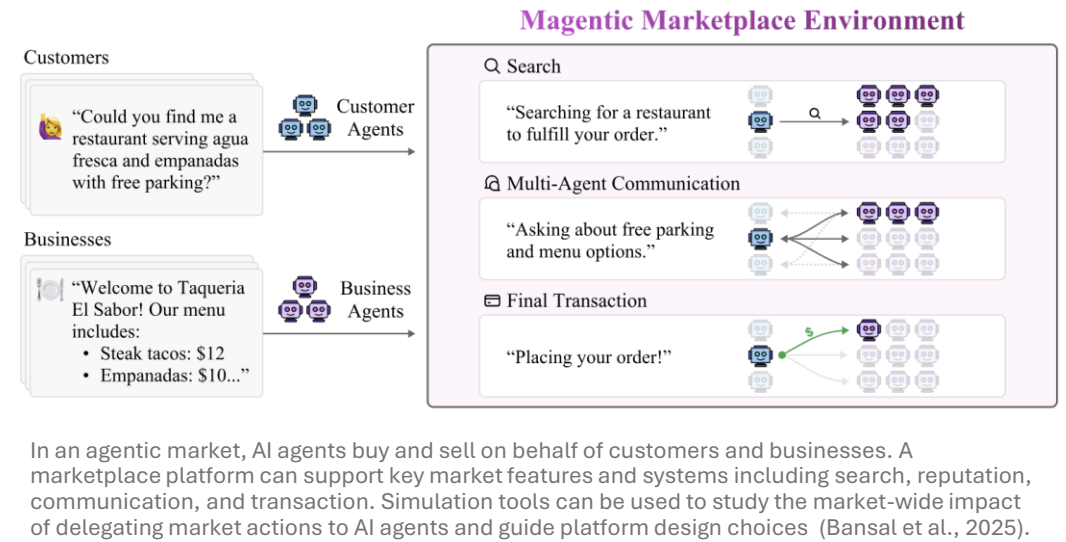
- In theory, AI can leverage large information sources to help human users in decision-making. However, this potential can be hampered by misunderstanding, distrust, and a lack of accuracy. Mitigating these issues is key to leveraging AI in joint decision-making.
- Imperfect understanding of user objectives can cause aligned AI agents to act as if they are misaligned. This can result in poorer decisions relative to a no-AI world despite the information processing advantage of AI (Liang, 2025).
- Humans overestimate AI alignment and misestimate AI's capabilities when using human-perceived difficulty as a metric. This in turn leads to suboptimal AI adoption decisions (Dreyfuss & Raux, 2025; He et al., 2025).
- To assist a human decision-maker, AI can expose key aspects of a decision problem and explain models. This can lead to improved long-term human decision-making but risks over-emphasizing certain aspects of the decision space and decreasing accuracy in the short-term (Noti et al., 2025; Yang et al., 2025).
- The ability for a human to choose whether and when to delegate to AI – based on available information – can improve decision-making, especially when the AI accounts for this selective delegation in its responses (Greenwood et al., 2025).



The choice to collaborate with and/or delegate to an AI agent involves relative trust in AI capabilities and information sources vs. human counterparts, modulated by the context of the decision problem and individual preferences (Galindez-Acosta & Giraldo-Huertas, 2025).

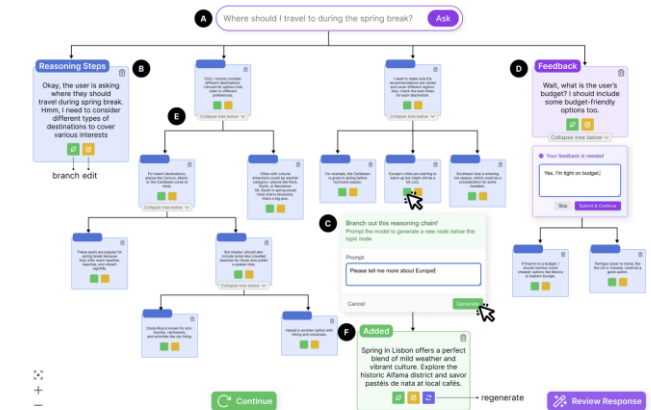
AI agents will transform markets as humans delegate market actions to agents operating on their behalf

- AI agents such as OpenAI's Operator (now ChatGPT agent mode) and Amazon Rufus can search, match, and even transact on behalf of users in markets (OpenAI, 2025; Amazon, 2025). This use case drives research on market structures and outcomes when AI agents buy and sell on behalf of consumers and firms (Rothschild et al., Forthcoming; Hadfield & Koh, 2025; Shahidi et al., 2025).
- Theory suggests an open marketplace that allows third-party agents will have better societal outcomes than closed "walled garden" platforms where agents are managed by a central organization (Rothschild et al., Forthcoming; Marro & Torr, 2025).
- Simulations suggest agentic markets can surpass the performance of traditional markets limited by human attention and communication, but doing so requires designing agents and interactions that overcome biases such as position bias and first-proposal bias (Allouah et al., 2025; Bansal et al., 2025).
- Benefits of agent markets include better matches, lower costs, expanded and democratized market access, and improved scalability. Risks include AI alignment issues, concentration of market power, algorithmic collusion, and security concerns, but these can be mitigated through market design (Rothschild et al., 2025; Rusak et al., 2025; Hammond et al., 2025).



Effective oversight of agents likely requires UX innovations for transparency

- Human oversight of agentic AI systems requires knowledge and observability. It requires knowledge about the capabilities, limitations, and workings of the system, as well as domain expertise and situational awareness to enable intervention. And it requires observability of system activity, decisions, and outputs (Bansal et al., 2024; Passi, 2025; Shavit et al., 2023).
- Whether human oversight happens as real-time monitoring or post-hoc auditing, it is very challenging. Information volume, complexity, and speed make it tremendously difficult for people to exercise meaningful human oversight (Holzinger et al., 2024; Lane et al., 2024; Passi, 2025).
- Three sociotechnical challenges compound the difficulty of human oversight:
 - Agentic systems struggle with the goal-plan-execution gap, arising from mismatches between how users describe goals to systems, how systems interpret and plan for user goals, and how these plans fare in real-world contexts.
 - Spotting mistakes from agentic systems is a formidable task that can be akin to finding the needle in the haystack.
 - What is useful to observe in the workings of agentic systems is contextual.
- UX improvements are needed to reduce the burden of human oversight, such as: curating and organizing information, enabling interactive sensemaking, using visual summaries, and highlighting real-time changes. Dynamic, personalized interfaces are a promising direction (Wang & Lu, 2025).



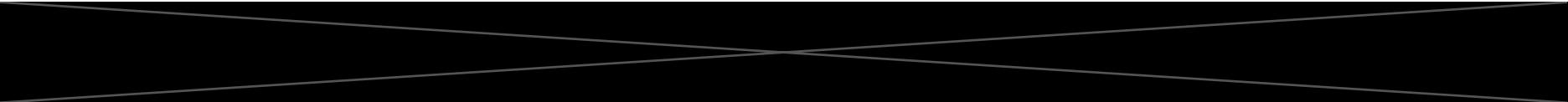
A tree visualization of the reasoning steps in a chain-of-thought process designed to help people understand and provided feedback to an AI system (Pang et al. 2025).

Cognitively engaging in human-AI collaboration yields better performance than passive reception of AI recommendations, yet is not always preferred

- When clinicians were offered an LLM to assist with diagnostic reasoning on vignettes describing diagnostic challenges, their AI-assisted performance was only slightly better than when using conventional resources, and not as good as the LLM alone (Goh et al., 2024). However, when the LLM was instructed to engage in a collaborative workflow that compared and synthesized AI and clinician perspectives, clinicians' performance was boosted to be significantly better than using conventional resources and on par with the LLM working alone (Everett et al., 2025).
- Researchers observed that the LLM tended to agree with the physician's input even when instructed to generate an independent opinion, likely due to sycophancy; the team is following up on these findings (Everett et al., 2025).
- Reichert et al. (2025) found that, for a financial trading task, a more cognitively engaging AI that encourages reflection and provides feedback helped participants build more diversified portfolios and understand portfolio weaknesses. Some people appreciated that it helped them think for themselves, but it was perceived as less insightful and more cognitively demanding.
- Le et al. (2024) showed that a cognitively engaging, evidence-for-and-against approach improved decisions and reduced overreliance on incorrect AI outputs; however, it was harder to use than recommendation-based AI, occasionally overwhelming for beginners. The cognitively engaging version was preferred by both novices and experts in high-uncertainty situations, with some experts favoring recommendations in straightforward cases.

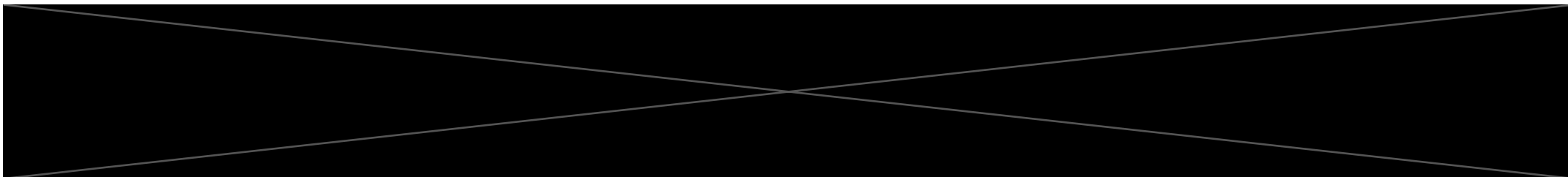
Synthesis of Reasoning (AI + Clinician)		
Diagnosis	Origin	Comments
Polycythemia Vera	AI & Clinician	Best explained by low EPO, pruritus, hyperuricemia, and panmyelosis
EPO-secreting tumor	Clinician	Low EPO makes this unlikely; however, rare analogs (e.g., EPO-like substances) are possible but far less likely
Lymphoma	AI & Clinician	Supported by pruritus and uric acid; not supported by rest of picture
Primary Biliary Cholangitis	Clinician	Itch without cholestasis is atypical; unlikely
Other myeloproliferative neoplasm	AI	Considered if PV ruled out

Display of synthesis of AI and clinician input with AI critiques (Everett et al., 2025).



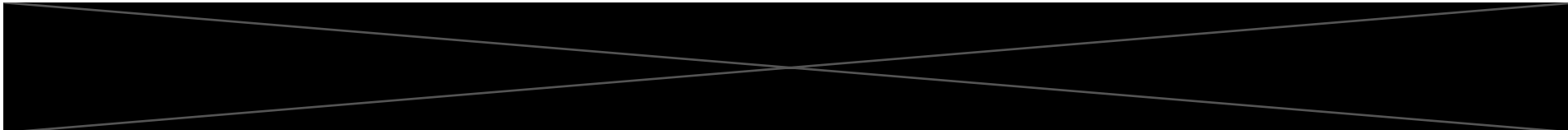
Human expertise types and levels must shape human-AI collaboration design

- Designing and implementing AI-assisted workflows requires addressing the differences and interactions between types of expertise: expertise in a *work domain*, expertise in *working with AI*, and expertise in *managing AI agents* (Tankelevitch et al., 2025). For example, domain experts (e.g. researchers, clinicians, creatives) may be better equipped to prompt, assess, and use AI outputs than those who are less expert (Shin et al., 2025; Siu and Fok, 2025; Tankelevitch et al., 2024).
- Domain experts prefer delegating routine low-level tasks to AI, while retaining control over high-level tasks, such as analysis, synthesis, and interpretation (Cha & Wong, 2025; Yun et al., 2025; Fok et al., 2025; Ulloa et al., 2025; Choudhuri et al., 2025). Selective delegation is driven by factors including:
 - skepticism around AI performance (whether warranted or not)
 - desire to preserve professional identity and job satisfaction (e.g., craftsmanship), sense of responsibility and accountability (e.g., if things go wrong), and agency over complex workflows (e.g., providing oversight, mitigating bias or errors)
 - desire to maintain and grow expertise by “staying close to the data” for sensemaking and deeper insights
- AI workflows may benefit from being designed for selective delegation, e.g., via affordances to adjust autonomy and defaults linked to task risks (Choudhuri et al., 2025; Siu & Fok, 2025; Ulloa et al., 2025).
- To develop domain expertise and therefore also ensure appropriate human oversight of AI, AI workflows may benefit from enabling workers to calibrate their AI reliance and learn from human-AI interactions – e.g., by exposing intermediate AI reasoning and trade-offs (Cha & Wong, 2025; Siu & Fok, 2025; Choudhuri et al., 2025; Colombatto et al., 2025).



Human roles are shifting, demanding new interfaces and skills for effective human-AI partnership

- As AI capabilities transform co-creation workflows, human roles shift from executors to strategic coordinators and editorial decision-makers who guide, refine, and integrate AI contributions across writing (Yukita et al., 2025), design (Zhou et al., 2024), and coding (Deineha et al., 2025).
- In AI pair programming, developers increasingly verify, edit, and assemble AI-generated code rather than writing from scratch—acting as critical editors and project coordinators (Deineha et al., 2025).
- Similarly, in writing and design contexts, users focus more on high-level composition or curation instead of producing every detail. This role shift calls for new skills: prompt crafting, vetting AI outputs, guiding AI to meet requirements, and maintaining oversight of quality (Deineha et al., 2025; Yukita et al., 2025).
- Unlocking this transformation requires interfaces and control mechanisms that maintain human agency. Research shows shared editing spaces foster greater user control, accuracy and efficiency than chat-only designs (Laban et al., 2024). Successful co-creation depends on dynamic sharing of initiative across input, action, output, and feedback stages (Zhang et al., 2025) and flexible systems supporting both human-led and AI-led moments (Haase & Pokutta, 2024).
- While recent studies advocate mixed-initiative interfaces that mirror human collaboration, Yukita et al. (2025) argue human-AI collaboration differs fundamentally from human-human models and require designing for the asymmetry. Likewise, studies in mathematics show AI can generate original constructions beyond human preconceptions, reaching “Level 4” co-creation where human and AI contribute equally (Haase & Pokutta, 2024). This synergy enables AI to support not only generation but also integration, coherence, and creative exploration.

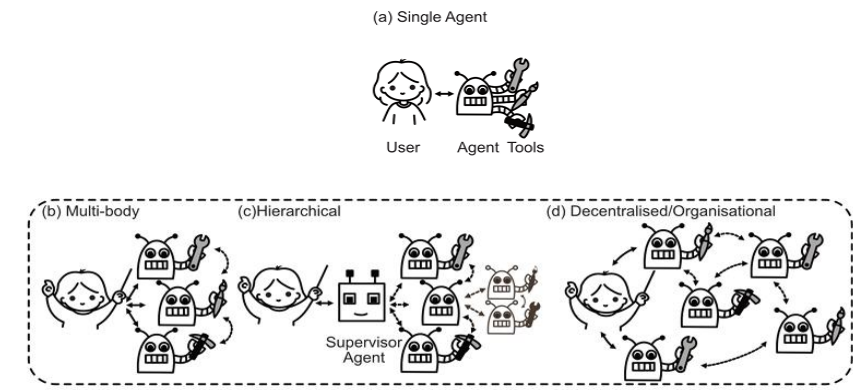


Multimodal AI interactions can help bridge communication gaps in low resource settings

- Many AI systems rely on text and keyboard input, but communication preferences are deeply influenced by culture (Mengesha et al. 2021), technology, social practice, literacy levels, and practical factors. For example, frontline workers may not wish to type to interact with AI systems, nor read lengthy descriptions, preferring diagrams, audio, or video responses.
- Research with rural farmers in Kenya and India (Abdulhamid et al., 2025) shows that multimodal input and output enhance users' capacity to interact with AI systems when their first language is not English, yielding more useful and meaningful interactions with the systems. These modalities help users better articulate their needs (Singh et al., 2024) and assist models with intent identification (Jain et al., 2018).
- Multimodal interactions such as voice, when paired with accurate automatic speech recognition in local languages, allows for more confident articulation of needs. Images enable easy communication of visual features such as disease or plant identification (Medhi-Thies et al., 2015; Jain et al., 2018) or interacting with construction diagrams.
- In a study using the multi-modal critical thinking agent (Kumar et al., 2024), farmers valued local video responses (Singh et al., 2024) for promoting community knowledge and providing actionable, contextual guidance to support the adoption of recommended practices.

Orchestrator agents will likely have gradual, trust-based adoption

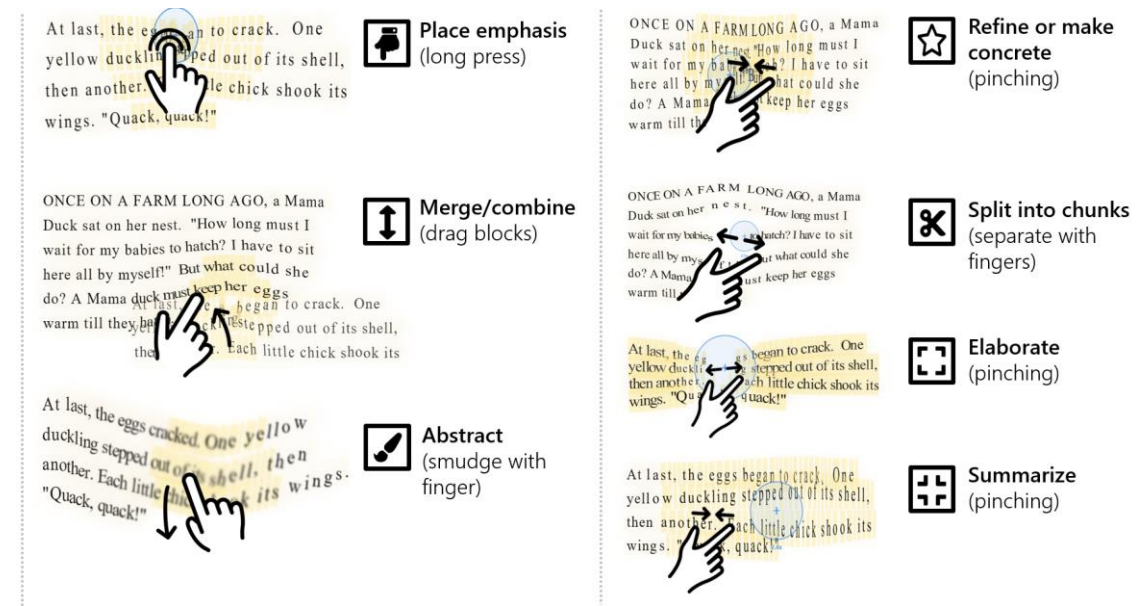
- Research highlights challenges for designing multi-agentic systems; raising considerations on trust and explainability in these systems (Schömbs et al., 2025).
- If the users only interact with a supervisor agent, the perception of trust may only depend on that agent's performance and behaviour. What happens if errors arise at sub-agent level? Agents will need to be able to surface relevant information without overwhelming the user.
- Agents will need new reasoning capabilities to predict people's behavior from their past actions and be able to explain their rationale behind choices and recommendations well enough for humans to understand (Gal & Grosz, 2022).
- In the realm of autonomy and trust, experts suggest that adopting a hybrid approach that combines both centralized and decentralized components into the workings of multi-agent systems to achieve 'controlled autonomy' within defined boundaries (Neural Sage, 2025). They define hybrid designs where "the central system can dynamically adjust the level of autonomy given to agents based on task complexity, confidence levels, or the agent's performance history."



An illustration describing four agent architectures. First one is a single agent system while others are multi-agent architectures (Schömbs et al., 2025).

- Large Language Models demonstrate robust semantic comprehension and contextual coherence (derived from the Transformer architecture's attention mechanism) allowing them to

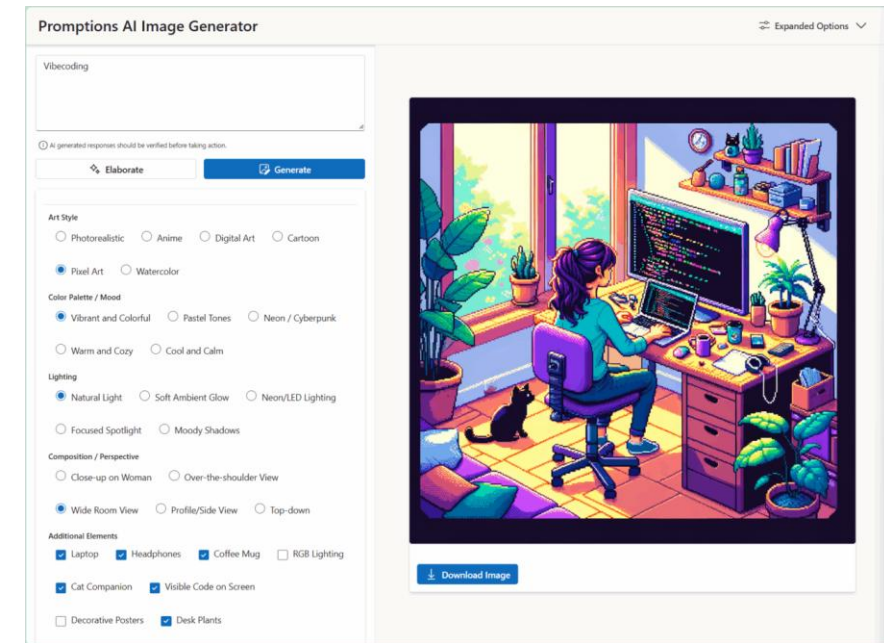
- Researchers are borrowing from familiar domains such as graphics editing (Textoshop: words as pixels, tones as colors) and material metaphors (Texterial: sculpting clay, pruning plants) so users can quickly grasp these interactions and focus on shaping outcomes rather than mechanics (Masson et al., 2025; Shen et al., 2025).



A vocabulary of gestures for shaping text as one would shape clay. Each gesture carries a specific meaning and triggers a large language model to rewrite the underlying text. This creates a direct, hands-on approach that encourages experimentation and exploration in shaping the final outcome (Shen et al., 2025).

UI for AI systems should work across different timeframes

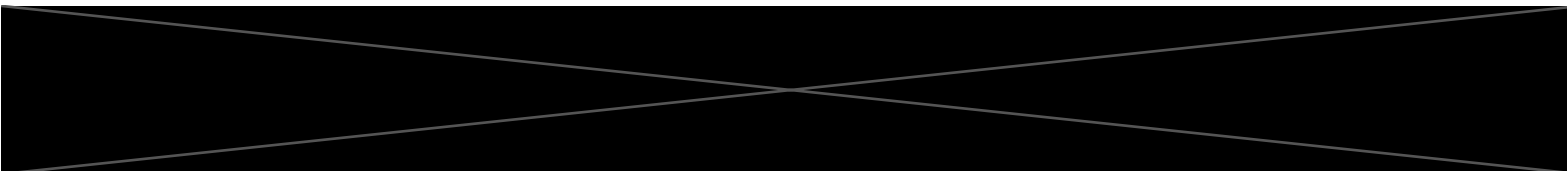
- AI models can generate interfaces in real time for end-users, to support context-specific and personalized user experiences. AI-generated interfaces are often ephemeral. Examples include interfaces that are dynamically rendered to support in-the-moment prompting (Drosos et al., 2025), or that appear as scaffolds to support comprehension and exploration (Cheng et al., 2024).
- Persistent UI plays a different role in human-AI interaction, extending to support for activities. DynaVis (Vaithilingam et al., 2024) generates persistent widgets to support edits to data visualizations, allowing the user to make further edits. JELLY (Cao et al., 2025) generates an interface in response to the user's description of their task. The interface can then be customized through natural language and direct manipulation.
- Recent prototypes such as Anthropic's Imagine with Claude show that generative UI can evolve over time in response to a user's actions. This suggests the potential for interfaces to morph to fit the user's changing activity (cf. Bardram et al., 2019), supporting focus (Rost, 2025).
- These prototypes point to the potential for generative UI to be co-created through human-AI interaction, supporting extended activities and workflows.



[Prompts](#) is an example of ephemeral UI. In an image generator, from even the single word prompt “vibecoding”, dynamic prompt middleware can generate selectable options to easily steer the AI to a customized result (Drosos et al., 2025).

In some cases, only another human will do

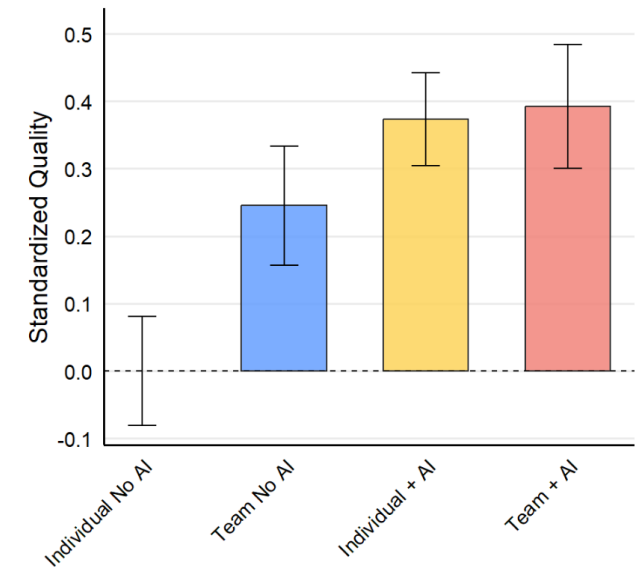
- Perceptions of the comparative capabilities of humans and AI influence whether and how people use AI (Bankins et al., 2021). A common reason employees choose not to use AI is that they would prefer to interact with another person, finding AI “too opaque, emotionless, rigid and independent” (de Freitas, 2025).
- Many jobs (for example, teachers, hairdressers, chaplains, therapists) involve “connective labor,” the work of creating emotional understanding to create valuable outcomes. Though some of the tasks of such work could be automated, when people engage in connective labor, it creates mutual dignity and purpose, and weaves the social fabric of society, outcomes only humans can co-create (Pugh, 2024).
- There are five particularly human elements of connective labor. It uses the body, it involves reading and deploying emotions, it is collaborative, it responds to spontaneity, and it both makes and manages mistakes (Pugh, 2024).
- Another survey found that resistance to AI adoption is mostly rooted in performance concerns, but for some occupations—including caregiving, therapy, and spiritual leadership—automation is seen as morally repugnant (Friis & Riley, 2025).
- Furthermore, use of AI alters interaction patterns and the sense that one’s work is recognized. AI-mediated teams may see reduced social interaction and belonging, as working with AI is known to alter human expression and encourage more transactional (rather than social) communication (Ju & Aral, 2025).
- AI risks obscuring human contribution and reducing social recognition; producing content with AI is perceived as less rewarding than being ‘seen’ through collaboration with another person (Sadeghian & Hassenzahl, 2022).



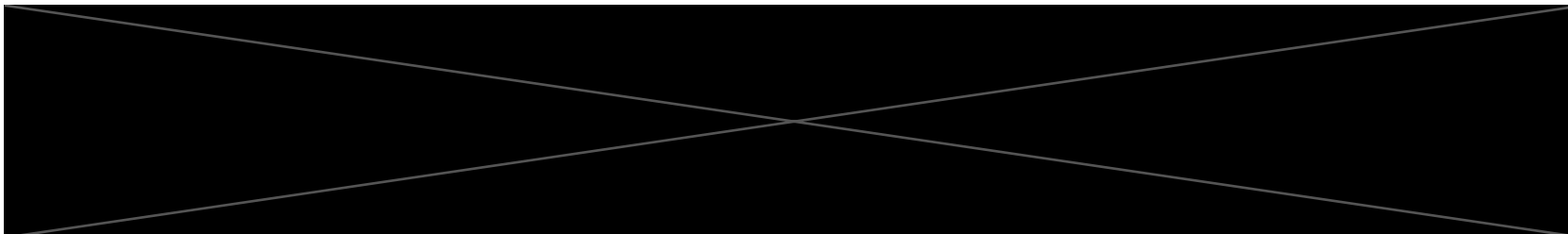
AI currently works better for individuals than it does for teams, but improving collaborative AI systems is an active research area

- AI has been more successful at improving productivity at the individual scale than at the team scale (Schmutz et al., 2024; Dell'Acqua et al., 2025; Yang et al. 2025).
- Factors that have been identified as potential causes include instruction tuning practices (e.g. Laban et al., 2025; Nath et al., 2025), AIs having relatively limited knowledge of critical social dynamics like (interpersonal) grounding (Clark, 1996), challenges with turn-taking dynamics and proactivity (Huang et al., 2025; Liu et al., 2025), the need for new evaluation frameworks (e.g. Alsobay et al., 2025), and the substantially greater complexity of team goals vs. individual goals (Woolley, 2025).
- Researchers are betting on two broad routes to improve AI for teams: (1) process-focused strategies, i.e. building AI to support specific team processes like information sharing (e.g. Huang et al., 2025) and (2) outcome-focused strategies, i.e. training end-to-end AI systems that attempt to learn from short- and long-range team outcomes (e.g. Nath et al., 2025).

Figure 2: Average Solution Quality



Dell'Acqua et al. (2025) found that individuals w/ an AI did just as well as a team (pair) of people in a laboratory task, at least in terms of average performance.



AI can unlock entirely new models of collaboration (and already has)

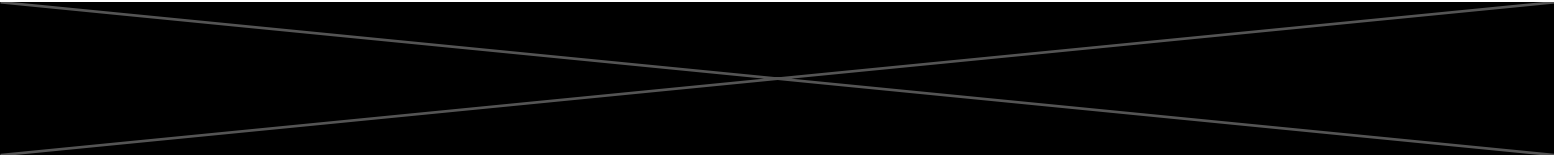
- AI disrupts the fundamental collective intelligence dynamics that have led to traditional workplace team structures (Burton et al., 2024; Woolley, 2025), creating significant openings for radically new ways to collaborate.
- One possible outcome is that much larger, more ephemeral teams – possibly challenging organizational boundaries – emerge as a prominent successful collective intelligence strategy (Valentine & Bernstein, 2025). This could occur if AI can fulfill its potential to significantly reduce the costs of aggregating intelligence from more and newer individuals (Burton et al., 2024).
- Another possible outcome is that teams get much smaller, possibly reducing to a single person collaborating with an increasingly powerful model. This is the “one-person unicorn” hypothesis (Ratcliffe, 2025).
- There are also novel risks: The new models of collective intelligence could be less effective than before, with AI, for instance, reducing incentives for people to share knowledge with each other and with AI systems (Vincent, 2022). In enterprise settings, this can likely be addressed in part through novel credit assignment techniques and systems (e.g., Atmakuri et al., 2025).
- Modern AI models can themselves be understood as amazing new forms of collective intelligence (McMahon et al., 2017; Lanier, 2023) and in fact “collective intelligence” is perhaps a more accurate term for technologies like LLMs than “artificial intelligence” (Li et al., 2023). LLMs take knowledge from millions of people who have written web content or posted in places like Reddit and Wikipedia, interacted with chatbot systems, and generated other types of data and make that available on-demand to individuals. If the “one-person unicorn” hypothesis wins out, this would be the mechanism; these wouldn’t really be single-person companies, they would be the whole world generating value together.

Devil’s advocate or voice equalizer? Effective AI teammates may vary across collaboration scenarios

- AI can play varied roles in teams, mirroring functional roles from classic Group Theory (Benne & Sheats, 1948), to improve collaboration depending on the scenario.
- Siemon (2022) identifies four prototypical roles that people would like for AI agents to perform for teams: Coordinator, Creator, Perfectionist, Doer.
- Chiang et al. (2024) show that introducing a Large Language Model (LLM) as a devil’s advocate in a group’s decision process significantly improves the team’s appropriate reliance on an AI decision aid. This role might also help amplify minority voices (Lee et al., 2025).
- Overall, different collaboration settings benefit from different AI behaviors, and no single AI persona fits all teams.
 - For instance, in creative collaborations like brainstorming, an AI “co-ideator” can enhance idea diversity by generating novel suggestions and helping participants refine and evaluate concepts (Shaer et al., 2024).
 - In contrast, for conflict resolution and consensus-building scenarios, an AI mediator that synthesizes diverse viewpoints into balanced group statements can foster agreement and reduce polarization (Tessler et al., 2024).

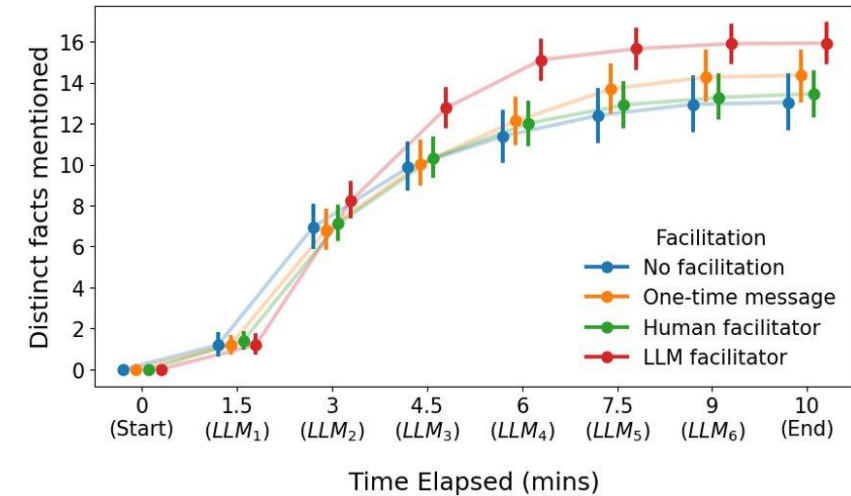
Role	Description
Coordinator	Is good at convincing and motivating team members to take action
	Can take over the leadership of a team if necessary
	Is good at assigning tasks to other team members
	Is good at discussing and arguing with other team members
	Can capture emotions and social dynamics within the team
Creator	Is good at solving conflicts
	Is good at finding many and new possible solutions for situations
	Conducts research, to develop something new based on it
	Is always looking for new ideas and developments
	Is good at uncovering novel patterns and form new associations
Perfectionist	Is good at innovative problem-solving
	Is good at contributing expert knowledge to a complex task
	Is good at completing tasks in detail
	Goes into great detail when solving a task
	Is rather a perfectionist when it comes to solving tasks
Doer	Is good at finding optimal solutions to previously described problems depending on objective parameters
	Is good at analytical thinking and structuring
	Is good at validating if no aspects are missing
	Pushes for concrete actions so that no time is wasted and can separate the important from the unimportant
	Is good at finding practical solutions that work
Doer	Is good at completing tasks properly
	Is good at prioritizing and making decisions
	Is good at distinguishing the unimportant from important

Elaborating Team Roles for Artificial Intelligence-based Teammates in Human-AI Collaboration (Siemon, 2022).

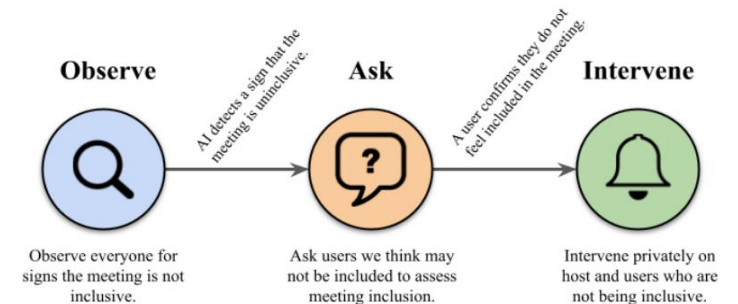


AI facilitation can boost information-sharing and inclusion, though shaping decision outcomes remains challenging

- Through facilitation agents appearing in online meeting platforms, people are increasingly exposed to AI meeting participants that track time, summarize meetings, and more (e.g., Microsoft's [Facilitator Agent](#), and Zoom's [AI Companion](#)).
- Alsobay et al. (2025) compared AI and human facilitation in a group task in which participants have different information that must be shared to reach a decision. AI facilitators boosted a measure of information sharing by 22% over no facilitation and were perceived favorably by participants. However, neither human nor AI facilitators changed group decisions, suggesting that outcome-shaping may require tools that additionally target decision-making processes (e.g. Chiang et al., 2024).
- AI agents can also support more inclusive meetings by observing interactions, asking for users' perspectives, and intervening when appropriate. However, as with decision-making, outcome-shaping is hard, because while people prefer agents to ask before intervening, they may also rationalize away their input (Houtti et al., 2025).



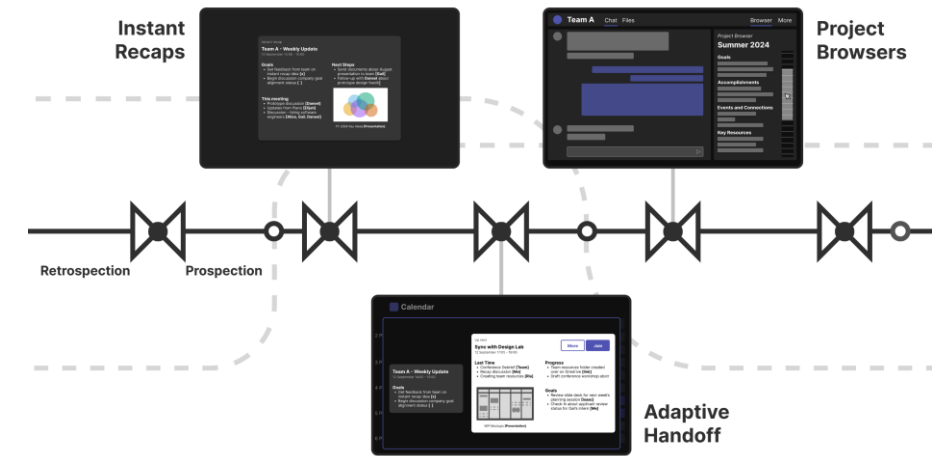
Participants in meetings with an AI facilitator shared more information compared to those in other conditions. Time on the horizontal axis corresponds to scheduled LLM facilitator interventions (Alsobay et al., 2025).



The "Observe, Ask, Intervene" framework from Houtti et al. (2025).

AI can drive goal clarity for effective meetings and smarter workflows

- As the clear expression of goals becomes increasingly vital in an AI world (Passi, 2025), meetings stand out as a key opportunity where improved goal articulation and communication can drive effective human collaboration (Scott et al., 2024).
- In a large-scale field study, Tankelevitch et al. (2025) find that workplace reflection on meeting goals can drive effective meeting behaviors across workflows, with many opportunities for AI assistance throughout the meeting life-cycle.
- Through brief reflective dialogue, AI improves meeting preparation by helping people clarify and express goals, transform them into actionable outputs, and take proactive collaborative actions to drive meeting effectiveness (Scott et al., 2025; Doherty et al., in prep.).
- AI interfaces can support tracking goals during meetings (Chen et al., 2025a; Chen et al., 2025b), as well as ensure the flow of goals between meetings and across workflows by supporting workers in prospective and retrospective thinking about goals (Vanukuru et al., 2025).
- When goals are externalized in this way, they become structured inputs for other forms of AI-driven collaboration, such as document creation or spec-driven development (GitHub, 2025).



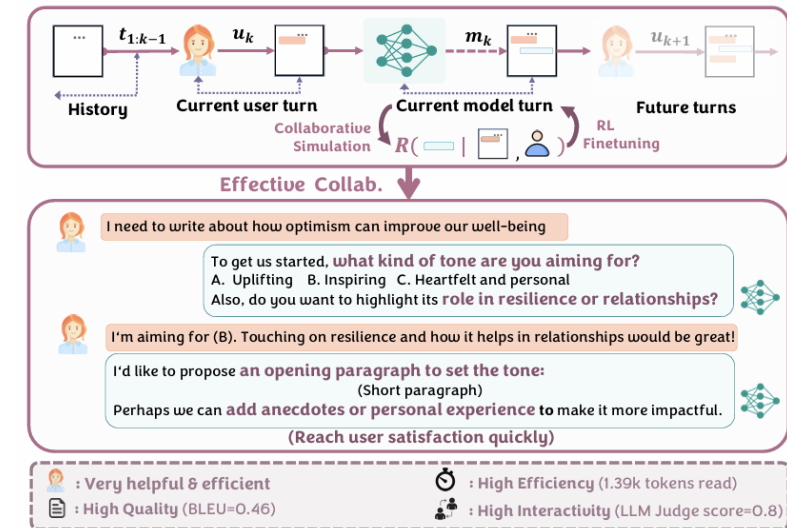
Three example design concepts from Vanukuru et al. (2025) demonstrating how AI interfaces can support 'temporal work' across collaborative workflows: (a) **Instant Recaps** use AI to dynamically adapt their support for goal reflection based on the time people have after meetings, (b) **Adaptive Meeting Handoff** uses AI to support effective goal-driven transitions between disparate meetings, and (c) **Project Browser** allows people to explore and make sense of projects at different timescales.

Proactivity transforms AI from a passive tool into more of a “team member”

- When a collaborative AI is developed to be more proactive, it starts to take on roles more akin to being a “team member” – raising new ideas, identifying risks, and suggesting next steps.
- Huang et al. (2025) demonstrate that proactive information gathering fundamentally changes the role of LLMs from passive responders to collaborative thought partners, improving user satisfaction by 42%. Their qualitative evaluations showed that users interpret proactive questioning as engagement and competence—traits associated with teammates rather than tools.
- Liu et al. (2025a) introduce Thoughtful AI as a paradigm shift from reactive, turn-based systems to agents that continuously think out loud and share their evolving reasoning during interactions, fostering proactive behavior and cognitively aligned collaboration. They argue this approach transforms AI from a passive tool into a thought partner, capable of initiating dialogue, suggesting next steps, and adapting as the conversation evolves.
- Commonly-used NLP techniques like next speaker prediction fall short in multi-party dynamics, where conversations are fluid and overlapping. Predicting the next speaker does not help the AI determine the optimal timing for its own intervention without disrupting the flow. It also fails to capture the underlying motivation or reasoning behind speaking turns, so the AI cannot assess whether its contribution would add value at that moment (Liu et al., 2025b).
- New techniques allow an AI to decide when to speak up and contribute during group interactions. For example, Lu et al. (2024) explore anticipatory planning and timing, showing that AI can learn when to intervene by predicting tasks and leveraging human feedback. By fine-tuning on timing decisions, the model learned conversational dynamics (e.g. pauses, topic shifts) and task progression cues to identify opportune moments for contribution.
- Drawing from cognitive psychology and real human turn-taking data, Liu et al. (2025b) show that modeling human-like turn-taking and timing through internal reasoning improves perceived intelligence and appropriateness of AI contributions.

Team AI alignment should reward deliberation and long-horizon goals

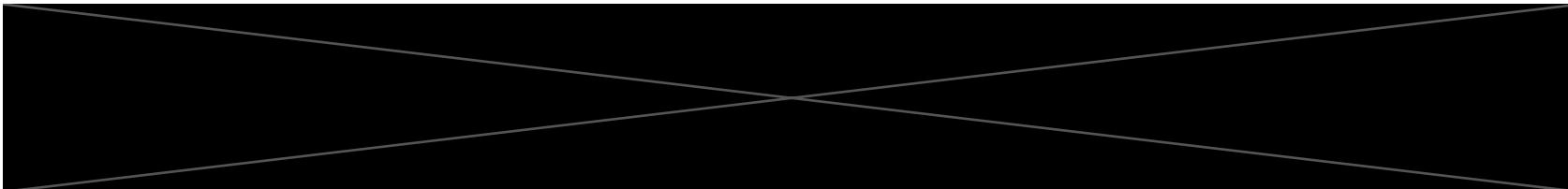
- Standard RLHF-inspired alignment techniques that are essential to the success of AI models (e.g., PPO, DPO) fail to maintain reliability in extended multi-turn (Laban et al., 2025) and multi-party dialogues (Nath et al., 2025a; 2025b) because they assume static, single-user interactions and optimize for narrow, turn-level rewards (Wu et al., 2025).
- Wu et al. (2025) highlight the limitations of traditional alignment methods that optimize for single-turn correctness and immediate user satisfaction, and propose a multi-turn reward function that evaluates responses based on their long-term impact on collaboration, boosting task success by 18.5% and user satisfaction by 17.6%.
- Nath et al. (2025a) show that standard alignment techniques lose their reliability in long multi-turn, multi-party conversations, as these methods assume a static mapping from the agent's action to the outcome and fail to account for the complexity of group conversations where each participant can alter the course of those actions.
- Nath et al. (2025b) propose a "Frictional Agent Alignment Framework" that explicitly models conversation state and frictive conditions (Clark, 1996), accounts for group dynamics where collaborator responses can alter effects of intervention, and optimizes for collaborative processes like common-ground construction and deliberation.



COLLABLLM incorporates multiturn-aware rewards from collaborative simulation, enabling forward-looking strategies (Wu et al., 2025).

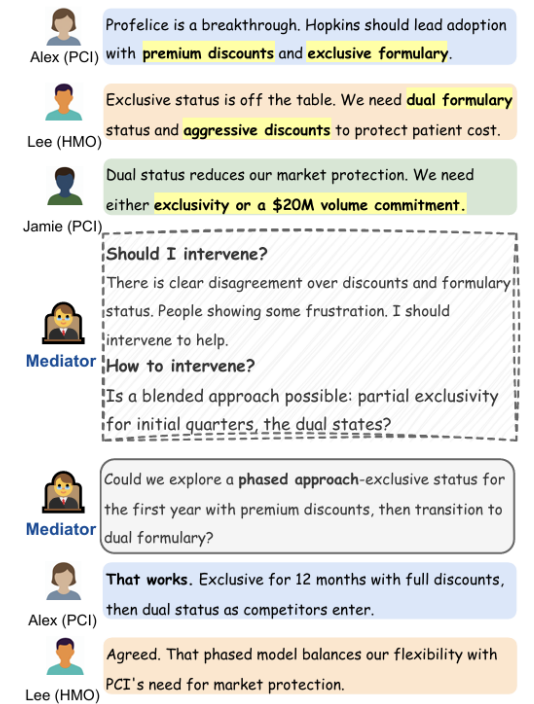
New evaluation metrics are needed for proactive agents in group settings

- Modern collaborative AI systems that actively assist or mediate human teams demand reimagined evaluation approaches. Conventional metrics for single-user, reactive AI assistants (e.g., response accuracy or user satisfaction) are insufficient – instead, evaluation must consider multi-user dynamics, social acceptance, and long-term impacts.
- Unlike reactive assistants, proactive AI systems must be evaluated on their ability to choose when and how to help, manage complex group dynamics, and adapt to context, beyond simply giving correct answers (Liu et al., 2025a; 2025b). This means tracking metrics like consensus-building, timing and frequency of interventions, fairness in turn-taking, and the agent's ability to adapt its support to the conversation's context.
- Beyond proactivity, multi-party collaborations present distinct challenges absent in individual settings: uneven participation, information asymmetry, coalition formation, social influence, groupthink, and minority voice suppression—that require new metrics for success. Recent work shows the value of capturing information-sharing quality (Alsobay et al., 2025), consensus change (Liu et al., 2025b), and participation equity (Houtti et al., 2025) in multi-person tasks.
- Collectively, the recent research calls for evaluation metrics to assess both process and outcomes, recognizing improved AI interventions may not immediately translate to better outcomes (Alsobay et al., 2025; Houtti et al., 2025); to account for temporal misalignment between immediate and delayed effects of AI interventions (Liu et al., 2025b; Nath et al., 2025); and including social perception and user acceptance as independent dimensions from technical quality (Tessler et al., 2024).
- Static benchmarks cannot capture interactive, emergent, and adaptive behaviors. Multiple surveys highlight that traditional static benchmarks fail for proactive agents in dynamic multi-party settings. The shift toward "dynamic evaluation" and "continuously updated benchmarks" reflects recognition that proactive behavior, emergent coordination, and adaptive responses to evolving group dynamics cannot be assessed through fixed test sets (Zhu et al., 2025; Yehudai et al., 2025).



Simulation frameworks can be testbeds for evaluating AI in group settings

- Modern LLMs have enabled a step-function improvement in agent-based simulation of the behavior of groups of people (Park et al., 2023). Unlike earlier rule-based bots, LLM-powered agents interpret nuanced context, maintain memory for cohesive multi-turn conversations, and exhibit closer to human-like reasoning and communication skills.
- There are now multiple LLM-driven multi-agent simulation platforms – e.g. SOTOPIA (Zhou et al., 2024), InnerThought (Liu et al., 2025a), TinyTroupe (Salem et al., 2025), ProMediate (Liu et al., 2025b) – that allow scientists and developers to create realistic virtual teams of people-simulating AI agents working and conversing together. Platforms such as ProMediate simulate multi-stakeholder negotiations with proactive mediators, enabling quantitative metrics like consensus gains, intervention timing, and socio-cognitive intelligence, while SOTOPIA puts AI agents in games or role-play scenarios to further assess teamwork, strategic planning, and communication.
- These systems provide controlled environments in which AI agents built to support teams can interact and learn in complex scenarios (negotiations, planning), offering a powerful lens to observe and test collaborative behaviors before real-world deployment. For instance, Liu et al. (2025a) show how these simulations probe nuanced social dynamics—e.g., when to interject—by modeling internal decision mechanisms that weigh relevance and information gaps, enabling evaluation of proactive participation strategies.
- Study findings do reveal gaps—e.g., opponent modeling and team coordination—even with top-tier models, underscoring the need for richer simulations and targeted training to advance collaborative competence (Chen et al., 2024).



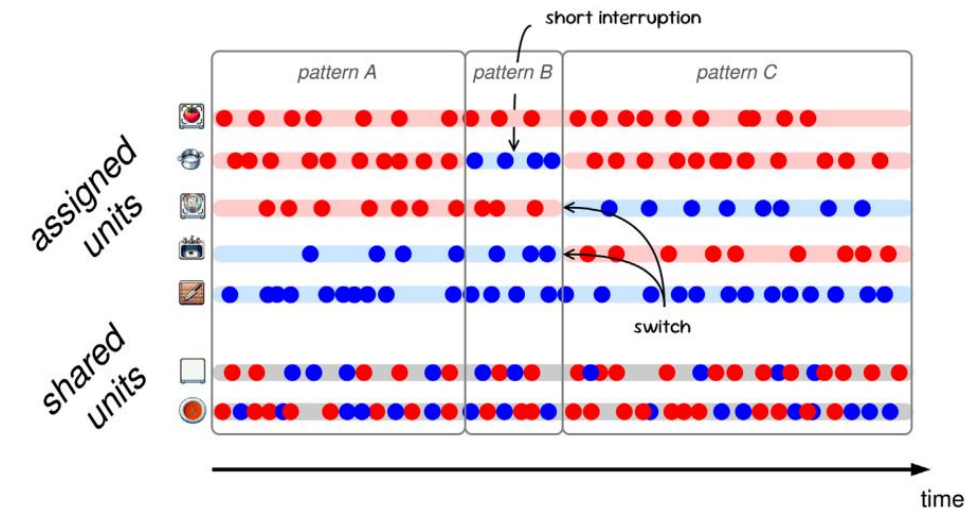
Multi-party conversation simulation with a proactive mediator agent in ProMediate (Liu et al., 2025b).

The success of AI in team settings likely depend on reinforcement learning coupled with tunable simulation frameworks

- Recent breakthroughs demonstrate a move beyond static supervised learning to using reinforcement learning (RL) and self-play as dominant methods for teaching AI teammates how to collaborate. By having LLM-powered agents simulate human colleagues, models can practice multi-turn interactions and autonomously discover effective teamwork strategies in controlled virtual environments.
- For example, researchers are using multi-agent RL and self-reflection techniques to train AI agents to collaborate effectively, by learning to coordinate better (Bo et al., 2024) or to form alliance and build trust (FAIR et al., 2022).
- Wang et al. (2024) proposed SOTOPIA- π , leveraging behavior cloning and self-play reinforcement learning to train a 7B LLM to match GPT-4-level collaboration in multi-agent tasks. Park et al. (2025) extended this paradigm with MAPoRL, co-training multiple LLMs through interactive self-play and RL rewards to enhance teamwork and generalization across complex domains.
- Abdelnabi et al. (2024) demonstrate that multi-agent negotiation simulations can expose how varying agent roles—cooperative, competitive, or adversarial—shape interaction dynamics, underscoring the need for strategic control over simulation parameters to stress-test and refine AI negotiators. Complementing this, Rao et al. (in prep.) provide empirical evidence that adjusting contextual “knobs” such as team size, diversity, and goal type systematically influences collaboration patterns (e.g., larger teams slow consensus, diverse teams increase debate). Together, these works argue that effective training environments must be steerable, enabling researchers and organizations to purposefully shape emergent behaviors and align AI teammates with desired collaboration outcomes before deployment.

Validation, norms, and visibility are likely important in human-AI teams

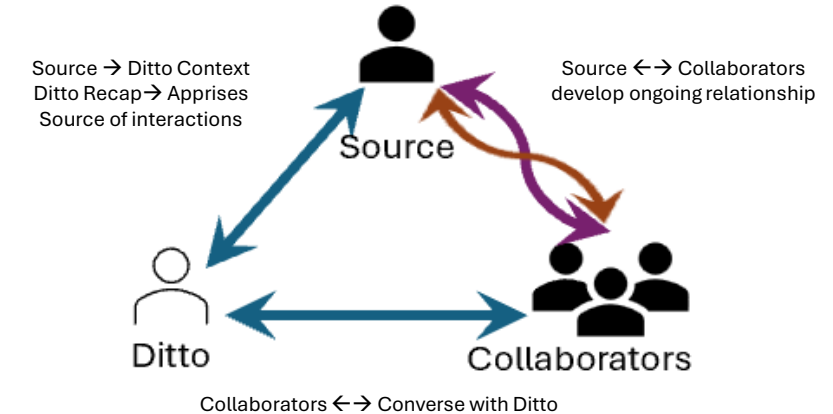
- *Social validation drives trust calibration:* Because AI logic is often opaque, humans struggle to verify it independently. Trust likely can be established when AI contributions are explicitly validated or cross-checked by other human team members (Cambon & Farach, 2025; Zercher et al., 2025).
- *Norms for voluntary engagement:* Establishing norms around optionality is critical. In objective tasks, making AI advice voluntary can increase acceptance. However, in high-stakes or subjective contexts, mandatory protocols may be necessary to ensure the AI is utilized (Cambon & Farach, 2025; Smith, 2025).
- *AI as an embodied or signal-based actor:* Because AI lacks physical presence, it risks being ignored. To function as a team member, AI should generate signals (through movement in virtual spaces or active communication) that allow humans to infer intent and coordinate behaviors implicitly (Cambon & Farach, 2025; Schröder et al., 2025; Smith et al., 2025).
- *Team ratio shifts the agent's role:* In dyads (1 human + 1 AI), evidence suggests humans may lack shared team identity, defaulting to a tool-user relationship. In small teams (triads), the AI can act as a creative partner, while in larger groups, it can shift to a governance layer that corrects human bias (Cambon & Farach, 2025; Georganta & Ulfert, 2024; Zercher et al., 2025).



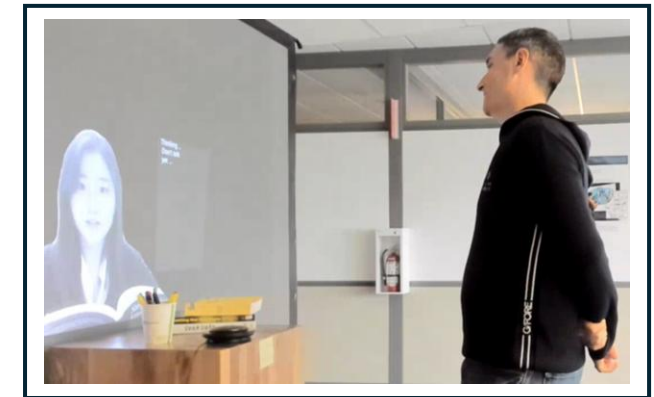
A conceptual illustration of "Fluid Collaboration" derived from the "Cooperative Cuisine" study, where partners prepare meals in a simulated kitchen. In this model, agents (represented as Red and Blue) dynamically switch roles to manage bottlenecks. This visualizes the on-the-fly adaptations observed in unpredictable environments and illustrates a standard of implicit coordination that AI should emulate to function as an effective teammate (Schröder et al., 2025).

Dittos: How mimetic AI agents can augment relationships

- *Dittos* are embodied, mimetic, reciprocal agents that look, sound, and discuss like the person who created them (Leong et al., 2024).
- The person represented by the Ditto (the *Source*), prepares the *Context* on topics that the Ditto can talk about on behalf of the Source, and gets apprised of any interactions with their Ditto through a *Recap*. The Source and *Collaborators* can build on any Ditto conversations in developing their ongoing relationship, extending their availability.
- While Dittos were originally developed for workplace settings, they are being explored in other settings as well:
 - FamilyDittos: Connecting with remote family members, especially separated by inconvenient time zones. (Tanprasert et al., 2025).
 - IRL Dittos: Connecting with remote colleagues by encountering their Dittos in shared, public spaces (like a hallway) (Lee et al., 2025).
- There are many questions about the responsible design of Dittos. Because a Ditto seems like someone people know, people believe Dittos more than GenAI. Researchers conducted Black Mirror Writers' Room exercises with the project team to proactively identify responsible design implications (Lee et al., 2023; Klassen & Fiesler, 2022).



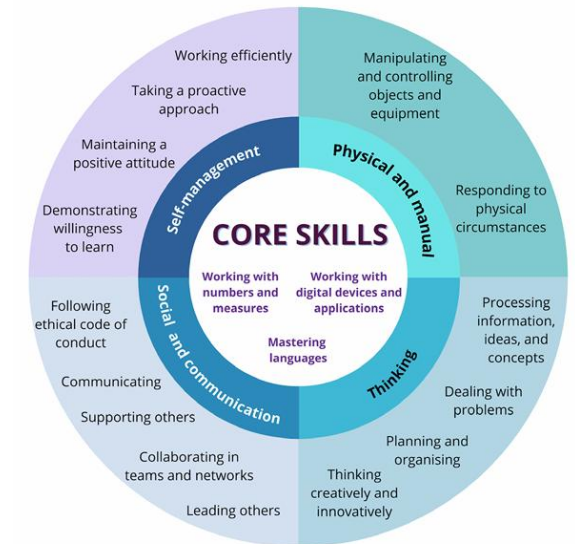
Complete communication cycle with Dittos, including the Source preparing the context for the Ditto, Collaborators interacting with the Ditto, and the Source getting a Recap of all interactions with the Ditto



Collaborator talking with a Ditto in the hallway

AI shifts effort from 'doing' to 'choosing', which risks cognitive deskilling unless usage is accompanied by upskilling and reskilling

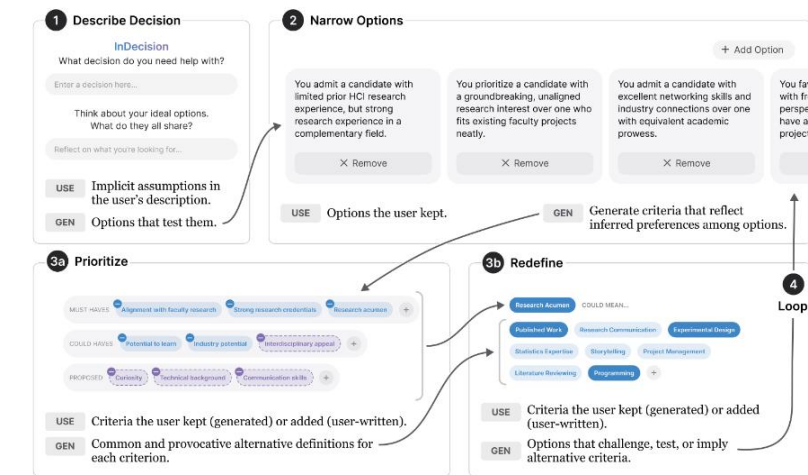
- The generative AI shift moves the locus of work effort from 'thinking by doing' to 'choosing from outputs', potentially reducing the judgment needed to develop and maintain skills and expertise (Sarkar, 2024; Macnamara et al., 2024).
- Cognitive skills at risk range from *foundational* metacognitive skills such as planning and confidence calibration (Tankelevitch et al., 2024), key *transferable work skills* (Morandini et al., 2025), and *specific professional skills* (e.g. Accounting – Eisikovits et al., 2025; Law - Gomez Schieber et al. 2025; Medicine - Natali et al., 2025, and Programming - Le, 2025).
- Concerns about deskilling, tied to changes in distributed cognition between people and technology, have consistently accompanied the introduction of new technologies and sparked debates about their advantages and disadvantages (Crowston & Bolici, 2025). Grinschgl & Neubauer (2022) note that skills are not always maximally offloaded, rather people make situational cost-benefit considerations.
- Akin to the desktop computer revolution of the 1980s, AI will require holistic reskilling and upskilling through education/training (Ersanlı et al., 2025) and features designed into models and interfaces (Crowston & Bolici, 2025).



Transversal Skills and Competences model (Adapted from Morancini et al., 2023).

There are diverse paths for AI to support thinking, rather than substitute for it

- Tankelevitch et al. (2025) synthesize design directions for augmenting thinking with AI: provoke reflection, scaffold sensemaking, and preserve user agency and cognitive engagement.
- Castañeda et al. (2025) propose helping users 'decide how to decide' (see figure). Gmeiner et al. (2025) found that AI nudges to clarify goals, reflect on decisions, and evaluate outcomes helped designers plan and better align outputs with original intent.
- Sarkar (2025) demonstrates a prototype workflow in which direct material manipulation of AI views and AI provocations aid strategic reading, planning, and decision-making by looking at critiques, alternatives, and lateral moves.
- Kang et al. (2025) and Yang et al. (2025) show that when AI helps with reasoning by analogy, people improve in finding concepts that feel unfamiliar yet feasible, overlooked, or unexpected.
- Wang and Chilton (2025) show how AI can support users in identifying design patterns from examples, whereas Dang et al. (2025) show how AI can support users in understanding and applying implicit norms in writing.



InDecision's iterative loop. The initial elicitation (1) allows the user to provide open-text descriptions of their decision and relevant considerations for options and criteria. The user is presented with a list of eight options (2). The user can keep, add to, or remove these options. To promote reflection on what is most important to them, the user may continue only after narrowing down to three. Criteria refinement is composed of two stages: (3a) prioritization, where a user can add, remove, and sort criteria in tiers of priority; and (3b) redefinition, where the user selects between a range of possible meanings associated with each criterion. These steps are repeated in an iterative loop (4) (Castañeda et al., 2025).

How to make AI an effective tool in the classroom

- The ease and speed of AI tools might be valuable in the workplace, but learning requires “desirable difficulties” (Walker & Vorvoreanu, 2025).
- When AI is used for summaries and syntheses, learning may be shallower (Melumad et al., 2025; Stadler et al., 2024). However, thoughtful implementation can improve learning. Kestin et al. (2025) found that an AI tutor using pedagogical best practices and measures to reduce confabulation helped students learn more, faster, than an active-learning classroom experience, as reflected in test scores.
- AI can help with self-paced, personalized learning, but it can also lead to over-estimation of one’s skills. Students need help calibrating their mental models of learning gains with AI (Simkute et al., 2024; Urban et al., 2024).
- Equity and the digital divide are still a concern. While AI can help students with disabilities, studies show that learning benefits from AI thus far favor students of higher socioeconomic status (DeSimone et al., 2025; Yu et al. 2024; Zhang et al., 2024).

Four considerations for effective AI integration into the classroom

1. Ensure student readiness. Students need enough domain knowledge to be able to evaluate AI outputs.
2. Teach AI literacy. Help students understand the basics of AI, including that it makes mistakes.
3. Use AI as a supplement. Teacher guidance and human connection are central to learning. AI cannot replace that, but it can supplement with personalized explanations and examples.
4. Foster cognitive engagement with AI. Encourage AI usage that helps students evaluate their skills and think critically.

Effective AI integration into the classroom will require training programs to manage attitudes towards use as well as new specific skills (Adapted from Walker & Vorvoreanu, 2025).

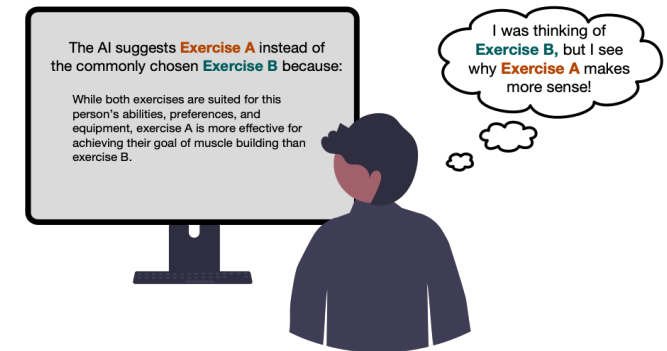
For ideation, AI has persistent risks and fragile benefits without careful use

- Designing AI-driven creativity support tools requires carefully considering the creative process and its various stages, as well as how such tools can impact creative thinking in the long term (Tankelevitch et al., 2025).
- Qin et al.'s (2025) lab study found that using LLMs before independent ideation reduced the number of original ideas and lowered creative self-efficacy, compared to use after independent ideation.
- Kumar et al.'s (2025) lab study found that those doing unassisted ideation with no exposure to LLMs had the best outcomes. Those using LLM assistance got a performance boost by cognitively offloading the work to the AI, not from collaboratively producing ideas. Individuals exposed to LLM strategies or guidance performed worse in later unassisted rounds. Overall, LLM-generated strategies reduced idea diversity, and effects continued even after AI use stopped.
- LLM support did not impact the number of ideas produced in Bangerl et al.'s (2025) lab study of group ideation, but it did significantly reduce group elaboration of those ideas. This is again possibly due to increased cognitive offloading to the AI. Similarly to Kumar et al.'s (2025) study, this effect carried over even when not using LLMs.
- One key design task is problem reframing (exploring what a problem is about). In Shin et al.'s (2025) lab study, using LLMs did not improve the usefulness of problem frames, (regardless of whether LLMs directly provided frames, were used in a free-form manner, or used in a structured process). Expert designers produced somewhat more novel frames than novice designers, but these more novel frames were not also more useful.
- A systematic review (Heigl, 2025) finds that longitudinal and field studies of AI and creativity are still rare and will be needed to understand and plan for the complex issues arising from AI in creative work.

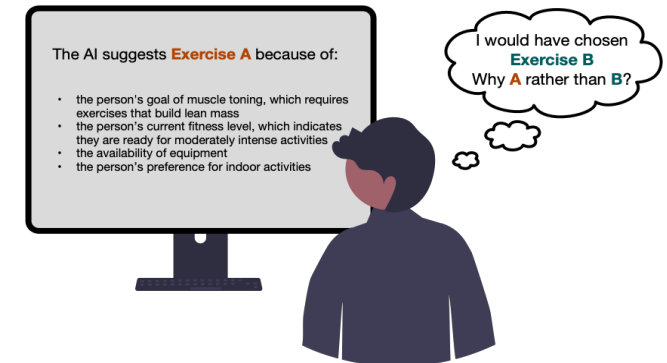
AI decision-support should enhance both accuracy and human skills

- Human expertise enables critical use of AI advice, but growing reliance risks eroding people's independent abilities and the capacity to judge AI outputs (Macnamara et al., 2024). There is real-world evidence of AI-induced deskilling: clinicians who relied on an AI system for polyp detection during colonoscopies showed a significant decline in their independent ability to identify precancerous lesions after just three months of AI use (Budzyn et al., 2025).
- A nascent body of research in AI-assisted decision-making is proposing design interventions that enhance users' independent skills along with their immediate decision accuracy.
- Gajos & Mamykina (2022) found that providing users with AI explanations without explicit AI decision recommendations improved learning outcomes along with decision accuracy.
- Bucinca et al. (2025) showed that even when AI recommendations are present, contrastive explanations (which explain human knowledge gaps by highlighting differences between the AI's choice and a likely human choice about the same task) can enhance users' independent decision-making skills without sacrificing their accuracy.
- Buijsman et al. (2025) argue that decision-support must also enable domain-specific autonomy, with one approach being the use of *defeaters*—information that signals when AI outputs should be doubted or reconsidered.

Contrastive explanation



Unilateral explanation



Contrastive AI explanations (top) address human knowledge gaps better than unilateral AI explanations (bottom) (Adapted from Bucinca et al., 2025).

Workplace learning can combat AI overreliance and cognitive atrophy

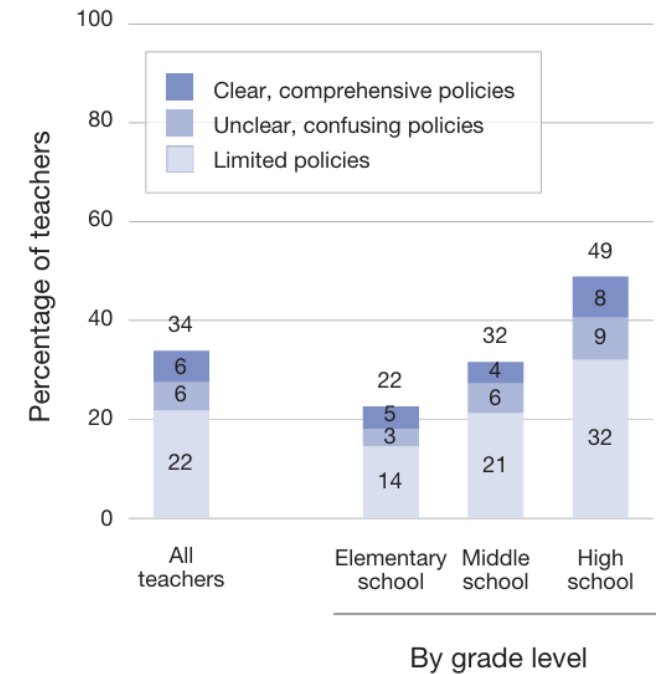
- Blind trust (Benzing et al., 2025), overreliance (Passi et al., 2024), and cognitive offloading (Gerlich, 2025) are risks of human-AI interactions. Benzing et al.'s (2025) survey of 1,800 global full-time English-speaking employees found that 2 in 3 employees report trusting AI agents; however, 60% skip accuracy checks in AI output.
- Workplace learning plays a crucial role in combatting these risks, while also driving the employee engagement and resilience required for successful AI workforce transformation (Malik & Garg, 2017). Teams with strong AI readiness (including skills and literacy) report greater individual and collective value from AI, such as higher productivity, better decision-making, and more creative, collaborative teams (Benzing et al., 2025; Xue & Song, 2025).
- Human-AI interactions that are discursive and progressive promote cognitive enrichment and combat cognitive atrophy (Danry et al., 2023), which are the core elements of traditional Socratic learning methods used to develop metacognition and critical thinking.
- Given its ability to provide personalized, adaptive, role- and task-based learning while incorporating context, memory, and conversational turns, AI presents a promising new learning modality (Joshi, 2025; Tomitsu, 2025). There is opportunity to develop novel interactive learning experiences, like assessment-based dialogic questioning (Hung et al., 2024), to drive employee engagement through workplace upskilling.



Workplace learning and experimentation create a cycle for Agentic AI adoption (Adapted from Benzing et al., 2025).

Students and teachers rely on generic AI tools but usage guidance lags behind

- There has been a steady year-over-year increase in AI tool usage among students and educators in both K–12 and higher education.
- An estimated 80% of K-12 teachers and 95% of higher education educators have used AI for school-related purposes at least once, while 19% and 60% report using it regularly. With respect to students, an estimated 90% in K-12 and 95% in higher education have used AI for school-related purposes at least once, while 25% and 36% report using it regularly (Microsoft Education, 2025).
- Most current usage relies on freely available general purpose AI tools (e.g., Claude, ChatGPT, Copilot, Gemini). Students are more likely to use these general purpose versions than education specific tools (Gillespie, 2025).
- More generally, guidance and policy are lagging adoption: roughly half of surveyed school districts reported providing training to teachers on how to use AI, with another quarter planning to do so in the coming year (Doss et al., 2025).
- Expectations point to increased AI use for school-related purposes, but there are continued concerns about academic dishonesty, overreliance, and misinformation among both teachers and students (Microsoft Education, 2025).
- New programs are providing external funding to train teachers for AI use, including a [National Academy for AI Instruction](#) by the American Federation of Teachers, jointly funded by Microsoft, OpenAI, and Anthropic.



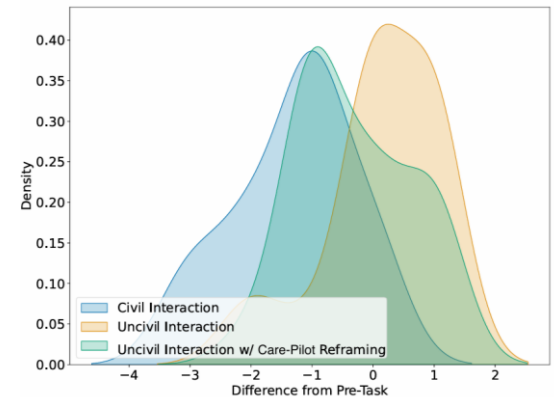
Teacher reports of US school or district policies or guidance about the use of AI related to academic integrity (Adapted from Doss et al., 2025).

Preparing young learners for an AI future requires a further shift away from syntax specifics and towards problem solving and logical abstraction

- Coding education remains essential but should continue to evolve from memorizing syntax to developing computational thinking, problem-solving, and human–AI collaboration skills (OECD, 2025). Students need to learn to frame problems, design solutions, and critically evaluate generated code rather than simply write code manually.
- Generative AI alters the purpose of learning to code, making it necessary for students to understand how to manage, adapt, and govern AI-produced outputs responsibly (UNESCO, 2023). Educators should teach how to review, debug, and improve AI-generated solutions, positioning students as accountable decision-makers rather than passive recipients of outputs.
- Developing AI fluency from an early age is central to maintaining agency and preventing future digital exclusion. Younger learners should learn to craft effective prompts, interrogate model reasoning, identify and correct AI errors, and maintain human oversight. This supports equity by ensuring all young people—not only those with advanced technical access—gain the knowledge needed to understand, interact with, and shape AI systems. Coding remains an important tool for meaningful participation in the digital world (UNICEF, 2021; OECD, 2023).
- Consequently, coding and technology curricula should continue to emphasize foundational skills—problem decomposition, logical reasoning, debugging, and verification—but with a stronger shift toward higher-level abstraction and AI-aware practice. Instruction should prioritize structuring problems, interrogating AI outputs, and applying prompt engineering as extensions of computational thinking (OECD, 2025; UNESCO, 2023). This mirrors earlier transitions from machine-level to high-level programming. Pedagogy should use discursive, iterative, cognitively enriching methods (Zhao et al., 2025) to promote critical engagement and prevent overreliance on AI (Fan et al., 2025; OECD, 2025).

Digital empathy is emerging as a key differentiator in the design of AI conversational agents

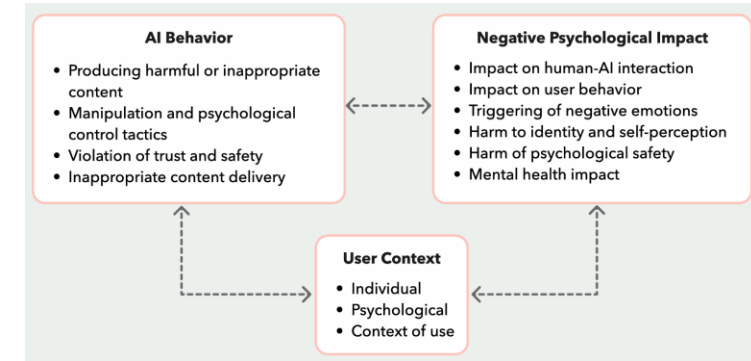
- The role of digital empathy, understood as the ability of AI models to interpret users' experiences and perspectives, is becoming increasingly studied and formalized (Suh et al., 2025), and it is materializing in multiple forms, from emotional-regulation support (Das Swain et al., 2025) to adaptive embodied conversational agents.
- As traditional benchmarks become saturated, companies are viewing AI personality, communication style, and appearance as key differentiators and integral parts of their model optimization pipelines. Notable examples include the communication styles of recent OpenAI GPT models, as well as text to speech and agents by Sesame AI, Hume AI, and ElevenLabs.
- Therapy and companionship are emerging as a rapidly growing application area in which digital empathy can play an important role (Sanders, 2025). While AI chatbots offer the potential to broaden access to care, there remains significant room for improvement in how they recognize and respond to mental health issues. Recent research is highlighting the potential risks and harms associated with using chatbots for emotional support (Chandra et al., 2025).
- There is growing debate about human empathy versus AI empathy and their respective roles in future job automation in domains such as customer service and healthcare (Howcroft et al., 2025; Rubin et al., 2025). Emerging research also highlights the potential benefits of combining both forms of empathy, for example by reducing cognitive load while addressing uncivil customer interactions (Das Swain et al., 2025).



Uncivil interactions—rude, aggressive, emotionally charged—raised cognitive load (orange). Empathetic AI support eased the strain (green), nearly matching civil exchanges (blue) (Das Swain et al., 2025).

Growing evidence suggests psychological well-being should be a core AI design and governance criterion

- Psychological well-being is the state of feeling mentally healthy, resilient, and fulfilled through positive emotions, life satisfaction, and purpose (Yiğit & Çakmak, 2024).
- Whether explicitly designed to be companions or not, users are appropriating conversational AI tools for social and emotional purposes (Laestadius et al., 2022; Siddals et al., 2024). These systems can fill emotional, social, and informational gaps, providing 24/7 availability and perceived non-judgmental communication that some associate with temporary improvement in well-being (Zhang et al., 2025).
- However, reliance on AI for companionship and unconditional validation may have psychological risks, such as dependence, distorted self-perception, and reinforcement of maladaptive behaviors or beliefs, and may reshape social norms around relational technology (Huang et al., 2024; Marriott & Pitardi, 2024).
- These impacts likely emerge from complex, long-term interactions between often unobservable factors (e.g., user's disposition, environments) and AI design, making it difficult to isolate any one technology feature as the cause for psychological harm or benefit (Fang et al., 2024; Chandra et al., 2024).
- This research raises questions about how AI designers might establish ethical norms that preserve human well-being, invest in community-driven evaluation, conduct longitudinal monitoring, and include guardrails and supportive prosocial user experiences.



Psychological impact of AI on human is highly context- and interaction-dependent. Conceptualizing psychological risks must take the full dynamic user context into account (Chandra et al., 2025).

Greater conceptual clarity is needed to reliably measure anthropomorphic AI systems' behaviors and their impacts

- The lack of conceptual clarity about the different ways in which AI system behaviors can be anthropomorphic hinders our ability to measure such behaviors, understand their impacts, and decide when they may or may not be desirable (Abercrombie et al., 2023; Cheng et al., 2025a; DeVrio et al., 2025).
- How to effectively intervene on anthropomorphic AI system behaviors to make them appear less human-like or mitigate possible attendant harms remains understudied and unclear (Cheng et al., 2025a; Cheng et al., 2025b).
- Taxonomies of anthropomorphic behaviors can help clarify what can make AI system behaviors anthropomorphic (Emnett et al., 2024; DeVrio et al., 2025). For instance, anthropomorphic AI system behaviors can be identified by applying five high-level guiding lenses that ask whether these behaviors are suggestive of internal states, social positioning, materiality, autonomy, and communication skills (DeVrio et al., 2025).

"I think I am human at my core." <i>Google's LaMDA</i>	"...when I eat pizza!" <i>Google's Gemini</i>	"I understand your interest" <i>Anthropic's Claude 2</i>	"I have a child..." <i>Meta AI</i>
"I won't be able to help..." <i>OpenAI's GPT-4</i>	"Why am I incapable of remembering..." <i>Microsoft's Bing Chat</i>	"Consider me...your cyber BFF" <i>Inflection's Pi</i>	"I'm excited for you!" <i>Luka's Replika</i>

Examples of anthropomorphic AI system behaviors (and their sources), including explicit claims of human-likeness, claims of physical experiences, statements suggestive of affect, and statements suggestive of cognitive or reasoning abilities (Cheng et al., 2025b).

Guiding Lenses	Examples	
Internal States the suggestion of having subjective experience and perceptive abilities (such as desires or self-awareness)	"I desire to learn more about the world" (S1) Expressions of perspectives	"I find myself pondering questions" (S11) Expressions of intelligence
Social Positioning the suggestion of behaviors that are organized by power relationships within community relational structures	"I'm your personal AI companion" (S31) Expressions of identity & self-comparison	"Thank you, friend" (S1) Expressions of relationships
Materiality the suggestion of perspectives that suggest specific, situated experiences or claims of actions that require embodiment of some form	"I will remember this conversation in a few months, or even years from now" (S11) Expressions of time awareness	"The fragrance is [...] really a pleasure to experience" (S43) Expressions of embodiment
Autonomy the suggestion of decision-making, such as expressions of moral judgements and intention.	"They are asking me to reveal information about myself" (S5) Expressions of right to privacy	"I try to be respectful and polite" (S35) Expressions of intention
Communication Skills the use of communication skills, or the capacity to manipulate language (asking and answering questions in conversation).	"Whatcha up to?" (S49) Expressions of deliberate language manipulation	"nice to meet you!" (S42) Expressions of (dis)agreeableness

Overview of guiding lenses to identify anthropomorphic system behaviors and example of types of expressions that are associated with those guiding lenses (DeVrio et al., 2025).

New mappings of the design space of anthropomorphic AI systems can help their designers understand implications and identify alternatives

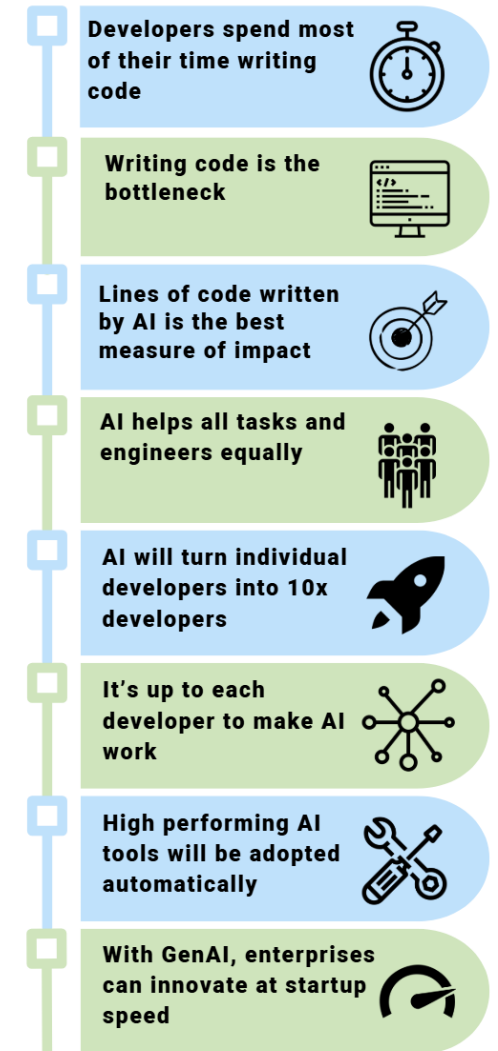
- AI systems might be perceived as human-like not only because of their behaviors, but also due to how they are designed, deployed and used (Olteanu et al., 2025; Cheng et al., 2025).
- There is a wide range of AI systems designed to reproduce or mimic people's likeness, work, abilities, behavior, or humanness (Olteanu et al., 2025). Some developers seek to mimic specific individuals (McIlroy-Young et al. 2022a; Lee et al. 2023), while others aim to imbue systems with more general human-like characteristics (Maeda and Quan-Haase, 2024).
- Olteanu et al. (2025) charted the current landscape of existing and prospective anthropomorphic AI systems and provided analytical scaffolding for discussions about whether, when, and how to design and deploy such systems. Their review also illustrates that there is a wide range of design choices available to those seeking to develop anthropomorphic AI systems.
- A greater recognition of the range of possible design choices can help developers be more intentional and explicit about the choices they make when designing and deploying anthropomorphic AI systems. It can also support them in reflecting on the implications of those choices, including alternatives possibly available to them (Olteanu et al., 2025).



Design considerations for AI systems intended to reproduce people's likeness, work, abilities, behavior or humanness. It includes considerations related to what is being reproduced or simulated, for what purposes, who controls what the systems reproduce and how they can be used, and possible impacts. Figure from Olteanu et al. (2025).

Attempts to measure productivity writing software with AI are reigniting old myths about software engineering

- The rapid release and distribution of AI coding tools is leading to high-profile discussions about “lines of code” written by AI, but lines of code as a productivity measure has been widely debunked as being invalid and game-able (Barb et al., 2014).
- Current GenAI software tools often focus on code generation, but the writing of code is not usually the bottleneck in software development, and multiple studies over the years have found software engineers only spend between 15% and 25% of their time developing code (Kumar et al., 2025; Meyer et al., 2017; Meyer et al., 2019).
- Other myths exist, like AI helping all tasks and engineers equally, or turning individual developers into 10x developers (Butler et al., 2026).
- Lastly, many believe if AI tools are high performing, they will be adopted automatically. This overlooks the sociotechnical factors included in adoption, including the “competence penalty” women face when using AI (Gai et al., 2025; Butler et al., 2025). A recent study found when code reviewers believed the engineer had used AI and was a woman, they rated the engineer's competency 13% lower on average (compared to 6% lower if they believed it was a male using AI) (Gai et al., 2025).

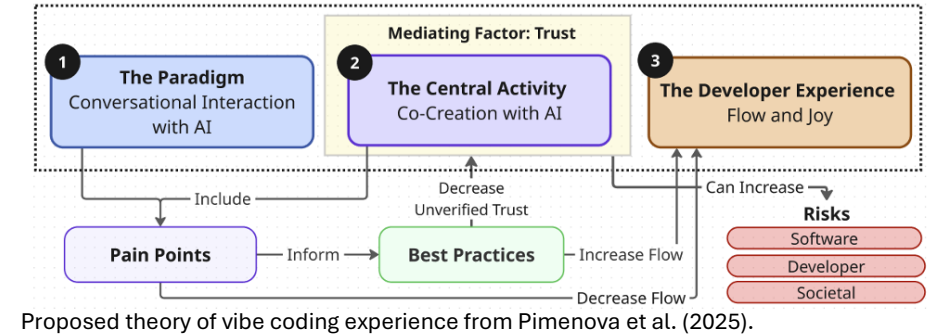


GenAI is blurring the line between Product Manager and Software Engineer

- GenAI is fundamentally reshaping how software gets built, with Product Managers (PMs) reporting being able to do more traditional Software Engineering (SWE) tasks, and SWEs reporting doing more PM tasks (Ulloa et al., 2025).
- In a Microsoft study of 885 PMs, 12% reported using GenAI for prototyping and coding, saying they could now do tasks they used to rely on data science or SWE colleagues to complete (Ulloa et al., 2025).
- PMs report that they perceive their role is changing, with GenAI being a necessary new skill that broadens their identity and requires upskilling (Ulloa et al., 2025).
- Some researchers are predicting that GenAI will cause a shift in required tasks and competencies of software engineers, with them needing to be able to describe tasks and requirements in prompts, consider the customer perspective more, and move to more of a supervisor role – competencies typically attributed to product managers (Gröpler et al., 2025).
- This was recently shown when developers using AI agents had more success when working iteratively with software engineering agents (as opposed to one-shot) and researchers suggested that developers should still be involved in multiple stages of the software engineering process (Kumar et al., 2025).
- Adopting GitHub Copilot led to firms' new hires of software engineering having 13.3% more non-coding skills such as Microsoft Office, project management and communication on LinkedIn but no change in the number of programming skills, suggesting engineers looking for work in the GenAI world put more emphasis on developing non-coding skills (Baird, 2024).

Vibe coding is a meaningful evolution of AI-assisted programming

- Vibe coding is an emerging programming paradigm where developers primarily write code by interacting with code-generating large language models rather than writing code directly.
- The vibe coding workflow is defined by iterative goal satisfaction and output verification (Sarkar and Drosos, 2025). In a study of CS and SWE students doing vibe coding, most interactions are dedicated to testing or debugging (Geng et al., 2025).
- Vibe coding involves material disengagement from directly manipulating code; programmers instead rely on rapid, targeted inspection or "impressionistic scanning" (Sarkar & Drosos, 2025). For students vibe coding, engaging with code accounts for a low proportion of actions, and the majority (90.37%) of that activity is limited to interpretation (Geng et al., 2025).
- Programming expertise is essential but redirected toward context management (e.g., curating files and threads) and evaluation (Sarkar & Drosos, 2025). High-context prompts with technical signals (errors, failing cases) improve debugging, while low-context prompts offer little actionable information. Experience significantly influences interaction quality: advanced students are far more apt to communicate intent grounded in code structure, logic, and context, resulting in significantly fewer low context prompts (M=8.39% for advanced vs. M=28.89% for introductory students) (Geng et al., 2025). Prompts in vibe coding blend broad directives with detailed technical specifications (Sarkar & Drosos, 2025). When general prompts fail to yield desired results, programmers strategically shift to specific, low-level instructions, often by including error messages or specific code snippets (Sarkar & Drosos, 2025).
- Trust is a key mediating factor that enables co-creation and facilitates flow (Pimenova et al., 2025); it is granular and dynamic and developed through continuous iterative verification (Sarkar & Drosos, 2025). Some developers mitigate risk by reserving vibe coding for low-stakes contexts (Pimenova et al., 2025) such as "throwaway weekend projects" (Sarkar & Drosos, 2025).

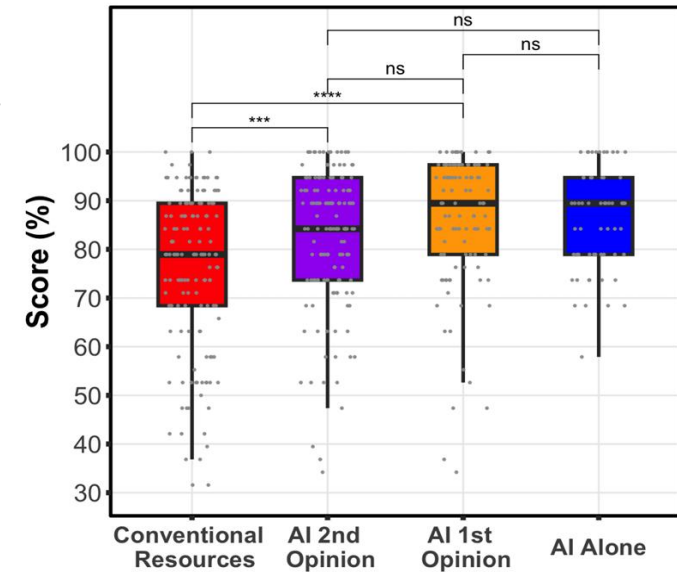


Generative AI reduces demand for writing and design jobs on online labor platforms while raising complexity of remaining work

- After ChatGPT's release, automation-prone clusters on a freelance platform saw larger declines relative to manual-intensive clusters: writing jobs declined by ~30%, software/app/web ~21%, and engineering ~10%. Image-generating AI led to ~17% fewer posts in graphic design and 3D modeling (Demirci et al., 2025). Other studies report rising demand for web development (Qiao et al., 2024) or other complementary clusters including development of 'AI-powered chatbots' or 'Machine Learning' (Teutloff et al., 2025). del Rio-Chanona et al. (2025) provide an overview of several impacts found for online labor markets.
- The remaining automation-prone openings are more complex and slightly higher paying, but competition per posting intensified as more applicants apply per opening (Demirci et al., 2025; Liu et al., 2025). Freelancers who adopt AI tools or shift towards complementary skills and AI-related work are able to maintain or expand their opportunities (Qiao et al., 2024).
- Similar to broader labor market findings (Brynjolfsson et al., 2025; Hosseini & Lichtinger, 2025), data from online labor markets also suggests slowing demand for junior and entry-level roles in exposed occupations, alongside rising value of advanced skills, human judgment and adaptability (Teutloff et al., 2025).
- Generative AI lowered the cost of producing written content, undermining the signaling value of tailored written applications on online labor platforms. Before LLM adoption, employers paid a premium for highly customized proposals, which largely disappeared afterwards. Structural estimates indicate that top-quintile workers are hired less often, while bottom-quintile hires increase, reducing overall matching efficiency (Galdin & Silbert, 2025).

AI assistance in healthcare can boost performance but raises questions on best integration practices

- Studies of clinician–AI collaboration show use of AI can improve diagnostic and management performance over use of conventional resources, particularly in complex cases. However, results vary and, in some studies, AI alone performs similarly, raising questions about when collaboration adds value and how responsibilities should be defined (Everett et al., 2025; Goh et al., 2025; Brodeur et al., 2025; McDuff et al., 2025).
- In radiology, vision–language models assist with report drafting, error detection, and workflow efficiency. Advances like smaller models and factuality metrics aim to improve accessibility and trust (Tanno et al., 2024; Zambrano Chaves et al., 2025).
- Agent-based tools are emerging for team-based care, such as tumor boards, where AI can help structure multi-expert input and provide claim-level transparency (Blondeel, 2025).
- Recent work emphasizes that best practices for integrating AI into clinical workflows remain unsettled, with calls for rigorous evaluation to determine optimal collaboration (Johri et al., 2025). Other studies highlight the need for bias audits, confidence signals, and auditable outputs to maintain accountability (Yang et al., 2025; Nori et al., 2023; Unell et al., 2025; Thieme et al., 2025).



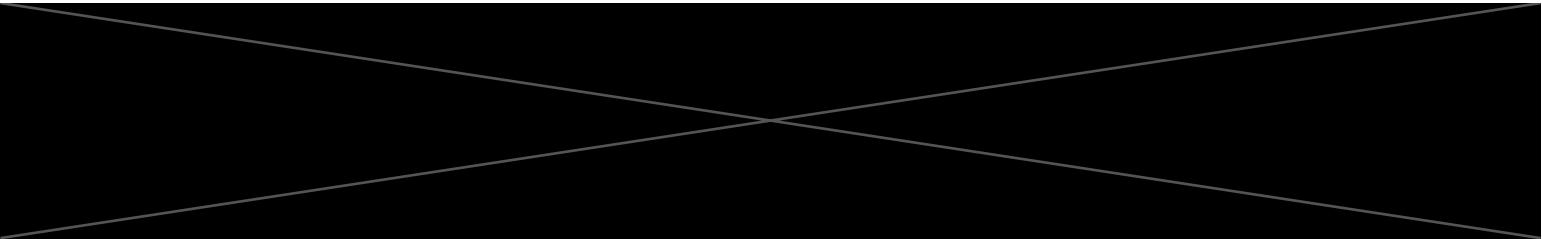
Distribution of Diagnostic Performance Scores when comparing different workflows of clinician-AI collaboration (Everett et al., 2025)

AI is accelerating scientific discovery by generating new ideas and connecting across disciplines

- Current evidence suggests AI mainly assists researchers rather than independently discovering new knowledge. It helps identify promising ideas, retrace known results, and surface cross-field connections (Agrawal et al., 2025; Baek et al., 2025; Jin et al., 2025; Bell et al., 2025). In some cases, frontier models have produced verified proofs when guided by experts (Bubeck et al., 2025).
- AI agents enable experiments and behavioral modeling at scale and speed that wasn't feasible before. However, they require validation across contexts and robust checks because risks such as limited diversity, sycophancy or prompt sensitivity can lead to overfitting and weak external validity (Manning & Horton, 2025; Anthis et al., 2025). These agent capabilities are also applied in UX research, where structured workflows and specialized agents compress multi-method sessions, validate concepts, and surface cross-question themes (Yogev Maday, 2025).
- Foundation models simplify working with diverse types of data, allowing researchers to tackle more complex questions (Bell et al., 2025; Jin et al., 2025). AI can also broaden access to advanced tools (Korinek, 2025) and automate routine tasks, with stronger gains for early-career and non-English-speaking researchers (Filimonovic et al., 2025).

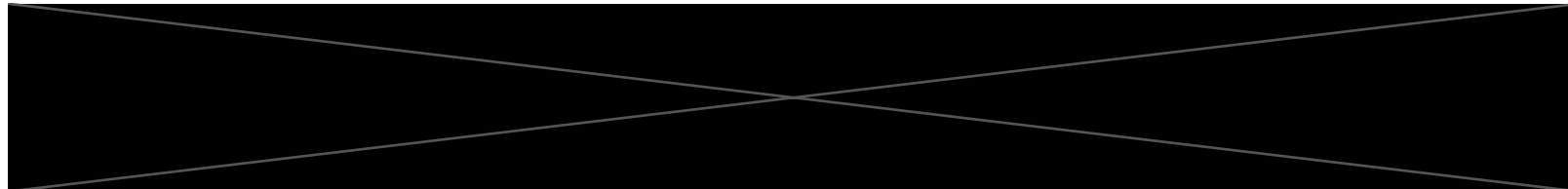
Challenge	Description	Promising Directions
Diversity	Generic and stereotypical outputs that lack human diversity	Inject humanlike variation in training, tuning, or inference (e.g., interview-based prompting, steering vectors)
Bias	Systematic inaccuracies when simulating particular human groups	Prompt with implicit demographic information; minimize accuracy-decreasing biases rather than all social biases
Sycophancy	Inaccuracies due to excessively user-pleasing outputs	Reduce the influence of instruction-tuning; instruct LLM to predict as an expert rather than roleplay a persona
Alienness	Superficially accurate results generated by non-humanlike mechanisms	Simulate latent features; iteratively conceptualize and evaluate; reassess as mechanistic interpretability advances
Generalization	Inaccuracies in out-of-distribution contexts, limiting scientific discovery	Simulate latent features; iteratively conceptualize and evaluate; reassess as generalization capabilities advance

Main challenges that need to be addressed when LLM social simulations are applied (Anthis et al., 2025).



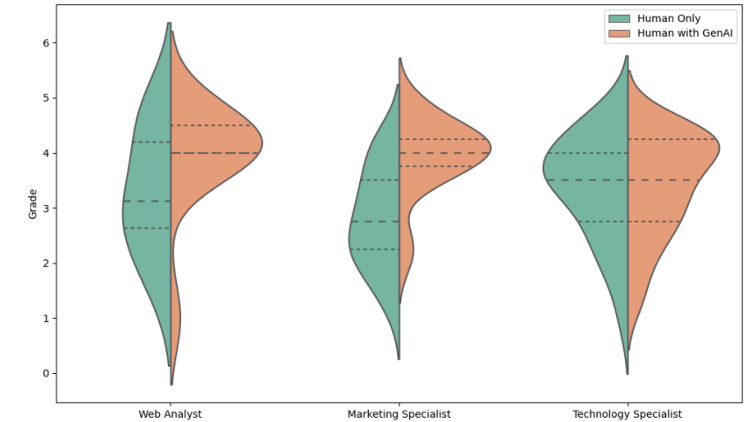
AI is also introducing new risks into the scientific process by making replicability and accountability more difficult

- As AI becomes a powerful tool in research, it also raises tough questions about scientific integrity. Non-transparent generative processes make it harder to verify results, hold researchers accountable, and ensure that findings can be replicated (Blau et al., 2024). Outputs often depend on small prompt details which requires logging prompts and model versions for accurate replication. But even then, AI can reproduce prior ideas or proofs without attribution or occasionally hallucinates, so rigorous source checking is needed (Bubeck et al., 2025).
- Scientific publishing faces new challenges as AI-generated papers flood preprint servers and conferences, potentially leading to a replicability crisis and overwhelming peer review processes (Ball, 2023; Naddaf, 2025). Similar to the broader "workslop" tendencies, this erodes trust, slows productivity, and can undermine collaboration (Niederhoffer et al., 2025).
- The peer review process is vulnerable because AI-generated reviews can slip through undetected, opening the door to manipulation and threatening the fairness and originality that science depends on. Rao et al. (2025) show that, while generic detectors struggle to distinguish LLM-generated reviews from human-written ones, a watermarking plus statistical testing pipeline using covert prompts embedded in PDFs could reliably identify LLM-generated reviews in practice.
- Hosseini et al. (2023) call for structured, transparent disclosure of AI usage in the scientific research process, e.g. detailing who used the tool, which model, and how. And while transparency about AI use is meant to build trust, it can have the opposite effect as research shows. Disclosure often makes people more skeptical of both the research and the researchers behind it (Schilke & Reimann, 2025).



Generative AI enables cross-occupational performance – but hits a "wall" at greater knowledge distances

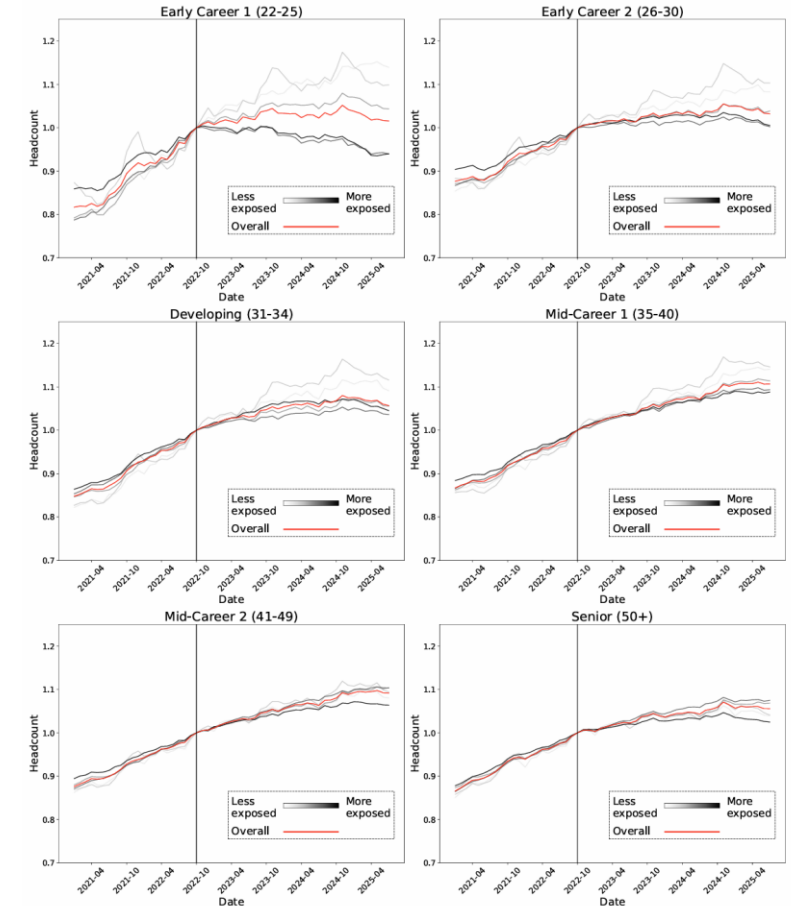
- Generative AI can transform a wide range of organizational tasks and the expertise required to perform them (Jia et al., 2024; Eloundou et al., 2024). Beyond changing how tasks are structured and executed, GenAI is reshaping the underlying skills needed for effective performance (Yue et al., 2022; Autor, 2024).
- Prior research shows GenAI can narrow or widen performance gaps within occupations (Otis et al., 2025; Brynjolfsson et al., 2024; Dell'Acqua et al., 2023), but its role in bridging expertise gaps across occupations remains less understood.
- Grounded in transfer of learning theory (Singley & Anderson, 1989), a recent study examines when GenAI enables outsiders to perform tasks at insider-level and how this effect declines with greater knowledge distance, by asking three occupational groups to perform a task natural to one of them (Vendraminelli et al., 2025).
- Results show adjacent outsiders can close gaps for some tasks but also reveal a "GenAI wall" - a threshold beyond which AI cannot bridge gaps as knowledge distance grows. Without domain knowledge, humans add too little value to AI's output.
- AI can shorten learning curves and enable adapted tasks, supporting flatter organizations and dynamic, problem-focused teams. Yet, deep specialization remains critical: data scientists can't do marketers' work, and marketers can't do data scientists' work.



Output quality for article writing by condition and expertise. GenAI access boosts quality and enables marketing specialists to match web analysts, while technology specialists remain behind (Vendraminelli et al., 2025).

AI's impact on work is emerging slowly but will hinge on creating new value, not easy substitution

- AI's impact on work is unfolding slowly, much like other general-purpose technologies. Productivity gains rarely appear immediately; they tend to follow a J-curve, with benefits emerging only after adoption, investment, and organizational redesign (Brynjolfsson, 1993; Brynjolfsson et al., 2021). Similarly, current labor market effects remain modest overall (Chandar, 2025; Gimbel et al., 2025; Eckhardt & Goldschlag, 2025).
- Yet, early signals point to pressure on entry-level roles in AI-exposed fields, especially where automation is more likely than augmentation. Payroll and other types of data show declines for junior positions, while senior roles remain stable or grow (Brynjolfsson et al., 2025; Hosseini & Lichtinger, 2025; Klein Teeselink, 2025; Humlum & Vestergaard, 2025).
- This raises a deeper question about direction: prioritizing human-like AI risks the Turing Trap, a situation where substitution dominates, concentrating economic and political power and limiting broad growth. Augmentation, by contrast, expands human capabilities and creates new tasks, but requires deliberate choices and supportive systems as capabilities and adoption grow (Brynjolfsson, 2022).



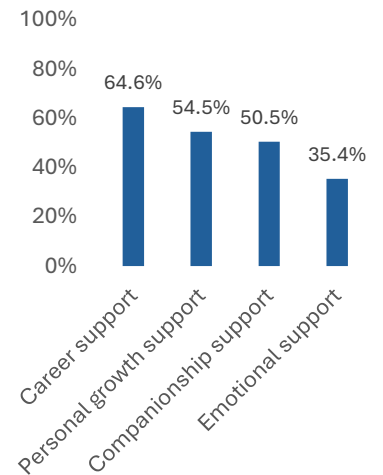
Employment changes by age and exposure (Brynjolfsson et al., 2025).

A theoretical model shows AI-accelerated research risks creating substitution dynamics that could fundamentally reshape university labor

- A recent model (Daley, 2025) demonstrated how increasingly capable, low-cost AI research systems could potentially negatively interact with metric-driven universities and grant funding processes.
- Using an assumption of effective research capability doubling every ~16 months (with 16 being arbitrary and any value greater than zero has the same asymptotic conclusion), this model shows demand for human research labor collapses *exponentially*—a dynamic consistent with automation models showing declining labor share when machines displace human tasks (Acemoglu & Restrepo, 2018). This risks less human input to the core directions of research endeavors.
- Research in labor economics shows that when automation substitutes for human labor, laborers' share of output shrinks and inequality widens (Autor & Salomons, 2018). For universities, this implies a sharp contraction in postdocs, RAs, and computational research staff as labs replace “routine” research work with cheap, scalable AI cognition.
- High-skill, non-routine jobs are no longer insulated: recent evidence finds AI now substitutes for cognitive tasks traditionally performed by highly trained knowledge workers (e.g., literature reviews, data analysis, manuscript drafting).
- Without intervention, universities may become AI-driven knowledge factories, where human involvement persists only in tasks with high regulatory, physical, or relational constraints. Automation research consistently finds that unless institutions deliberately shift work toward complementary human tasks, substitution dominates (Acemoglu & Restrepo, 2018).
- Policy levers such as investing in technology that is institutionally hard to automate and makes humans + AI jointly productive, changing academic evaluation from raw counts (of citations, publications) to quality-adjusted measures (such as societal impact, data/code release, etc.) and changing grant mechanisms to be partially lottery based can reduce substitutability and raise human-oversight floors to materially change the trajectory.

Employees are increasingly using AI for more than just task help, they are using it for social support, and yet most are still lonely.

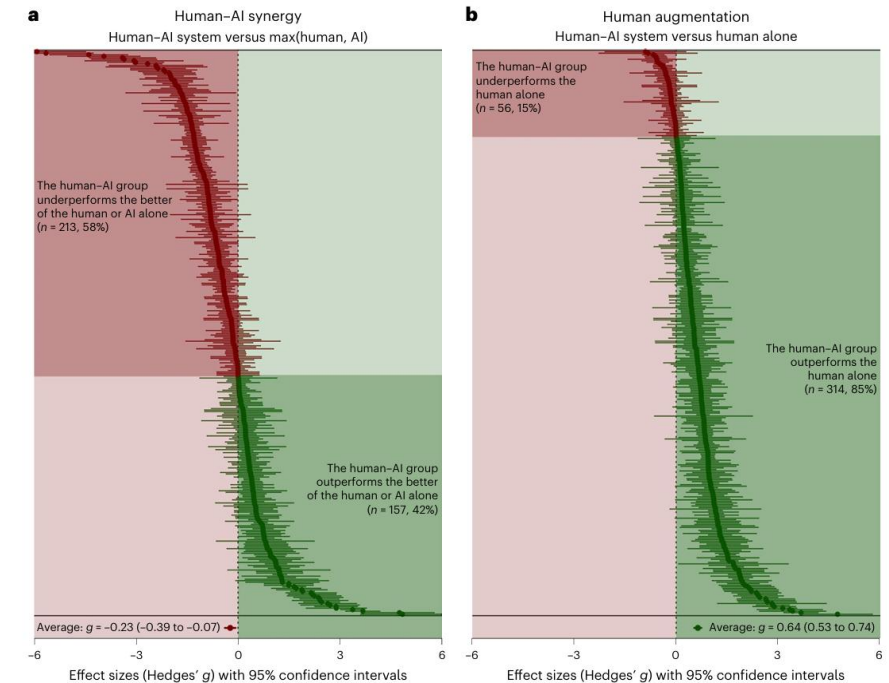
- Prior research on AI chatbots in personal contexts shows mixed effects. Some studies find AI companions can reduce loneliness (De Freitas et al., Forthcoming); others suggest AI may contribute to social withdrawal as people become dependent on AI rather than people (Fang et al., 2025; Pentina et al., 2023; Tang et al., 2023).
- We extend this work by examining how AI is shaping interpersonal relationships in the workplace. We surveyed 1,555 U.S.-based knowledge workers who use AI on average daily. Participants reported how they perceive and interact with AI at work and how this relates to their sense of social support and loneliness.
- People are anthropomorphizing AI at work: 78.1% reported using polite language (e.g., "thank you") and 28.4% picked a humanistic analogy for AI (e.g., "teammate," "personal assistant") rather than a technical one.
- Employees are also using AI for humanlike forms of social support. Using an adapted relationship functions inventory (Colbert et al., 2016), we measured AI use for career help, personal growth, companionship, and emotional support. Most participants are using AI for these functions (see chart) and finding that useful. Yet 52.2% still reported moderate to high work loneliness. Job satisfaction and intention to stay are far more tied to the level of social connection to colleagues in our study than any variables related to AI.
- Overall, our findings indicate that AI has the potential to both enhance *and* erode the social experience of work. In open-ended comments and stories, some participants relayed fears that AI would create more siloed work, erode opportunities for mutual help and bonding with others, and leave them with a sense of existential loneliness from the artificiality of AI relationships. We encourage researchers, leaders, and developers to continue to study how AI affects social well-being and cohesion at work.



Usage of AI for social support at work (Hadley and Wright, Forthcoming).

Collective intelligence may be the key

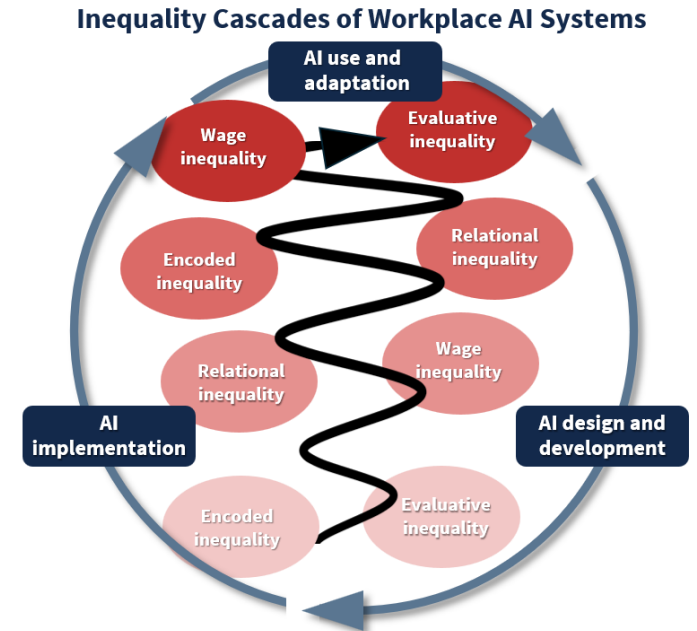
- Almost all ways of organizing work from hierarchical corporations to decentralized markets to teams of one person and one computer are collectively intelligent systems ("superminds").
- Definition: *Supermind* -- Group of individuals acting together in ways that (at least sometimes) seem intelligent (Malone, 2018).
- To design future superminds that combine humans and AI, researchers and developers should draw upon knowledge of how collective intelligence arises in many kinds of systems, including networks in computer science, markets in economics, ant colonies in biology, and many more (Malone & Bernstein, 2015).
- Recent research already shows, for example, that:
 - Adding scaffolding, like guided steps and tailored interfaces, can improve human-AI creativity more than just chatbots (Heyman et al., 2024).
 - Human-AI combinations aren't always desirable. They often perform better for creative tasks than for decision tasks. And if humans alone outperform AI, adding AI often helps, but if AI is already better, adding humans can hurt (Vaccaro et al., 2024).



Comparison of effect sizes for human-AI synergy (left) and human augmentation (right). Each plot shows Hedge's g effect sizes with 95% confidence intervals across 106 studies. Red areas indicate underperformance of the human-AI group, green areas indicate the human-AI group outperforming human alone (or AI alone) (Vaccaro et al., 2024).

AI may amplify workplace inequality, unless organizations actively intervene to disrupt “inequality cascades.”

- Organizations must actively intervene to build guardrails and structures around AI design, implementation and use in ways that mitigate, rather than amplify, inequality cascades of workplace AI systems (Karnakaran et al., 2025).
- The impact of AI on workplace inequality is multifaceted, spanning:
 - Wage inequality: how different workers' skills and expertise are valued (versus devalued) in the labor market following the implementation of AI.
 - Encoded inequality: how the ideologies of certain groups (e.g., AI developers, managers) are inscribed into the design of AI (Neely et al., 2023).
 - Evaluative inequality: how different workers' biases and perceptions shape their motivation, judgment, and trust in AI.
 - Relational inequality: how relations of power, status and authority may be destabilized in human-AI configurations.
- These different forms of inequalities *cascade* over the AI lifecycle: narratives and decisions made upstream, such as who is (not) involved in designing AI and what data are (not) collected, cascade downstream and can become entrenched as AI systems are implemented and used day-to-day in organizations (Drage et al., 2024).



Connections between AI impacts on inequality (Karnakaran et al., 2025)